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SUPERSEDING
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MILITARY SPECIFICATION

CONTROL ASSEMBLY, GYROSCOPE, ATTITUDE

A/A24G-39

This specification has been approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for an Attitude Gyroscope Control Assembly, designated A/A24G-39, which provides pitch, roll, and heading angular displacements.

1.2 Classification - The equipment controlled by this specification shall consist of the following components in accordance with 3.2.

Type Designation

Control Assembly, Gyroscope
Attitude A/A24G-39 (see 3.5)

Gyroscope, Displacement (see 3.6)

Amplifier, Electronic Control (see 3.6)

Controller, Compass System (see 3.8)

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in 3.2.2.

Beneficial comments (recommendations, additions, deletions, and any pertinent data which may be of use in improving this document) should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6610

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2. APPLICABLE DOCUMENTS

2.1 Is sues of Documents - The following documents of the issue in effect on date of incitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-D-1OOO	Drawings, Engineering and Associated Lists
MIL-W-5088	Wiring, Aerospace Vehicle
MIL-E-5400	Electronic Equipment, Airborne, General Specification For
MIL-C-5541	Chemical Conversion Coating on Aluminum and Aluminum Alloys
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel, Aircraft Equipment, Rack or Console Mounted
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification For
MIL-P-7788	panels, Information, Integrally Illuminated
MIL-M-7793	Meter, Time Totalizing
MIL-T-18303	Test Procedures, Reproduction, Acceptance, And Life For Aircraft Electronic Equipment, Format
MIL-N-18307	Nomenclature And Identification for Electronic, Aeronautical, and Aeronautical Support Equipment Including Ground Support Equipment
MIL-T-19576	Transmitter, Remote Compass, Thin Wing Type ML-1 (Unstabilized)
MIL-S-20708	Synchros, General Specification For
MIL-K-25049	Knob, Control, Equipment, Aircraft (Asg)
MIL-C-25050	Color, Aeronautical Lights and Lighting Equipment, General Requirements For
MIL-F-25173	Fastener, Control Panel, Aircraft Equipment
MIL-L-25467	Lighting, Integral, Red, Aircraft Instrument, General Specification For
MIL-C-26500	Connectors, General Purpose, Electrical Miniature, Circular Environment Resisting, General Specification For
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification For
MIL-D-38134	Detector, Magnetic, Azimuth DSU-4/A, High Temperature
MIL-C-81322	Grease, Aircraft, General Purpose, Wide Temperature Range, Barrier Coating Solution and Solvent
MIL-B-81744	Barrier Coating Solution, Lubricant Migration Deterring
ML-L-81846	Lubricating Oil, Instrument, Ball Bearing, High Flash Point

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STANDARDS

Federal

FED-STD-595 Federal Standard Colors

Military

MIL-STD-143 Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-454 Standard General Requirements for Electronic Equipment
MIL-STD-704 Aircraft Electric Power Characteristics
MIL-STD-781 Reliability Tests, Exponential Distribution
MIL-STD-794 Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810 Environmental Test Methods
MIL-STD-831 Test Reports, Preparation of
MIL-STD-889 Dissimilar Metals
MIL-STD-1334 Process for Barrier Coating of Anti-Friction Bearings
MIL-STD-2074 Failure Classification for Reliability Testing
MS17322 Meter, Time Totalizing Miniature, Digital 115 Volt 400 Cycle
MS21384 Knob Pointer, Illuminated and Non-illuminated
MS24264 Connector Receptacle, Electrical, Flange Mount, Miniature
MS24266 Connector, Plug, Electrical - Straight, Miniature
MS25164 Skirt, control Knob
MS25169 Knob, Bar, Small
MS33558 Numerals and Letters, Aircraft Instrument Dial, Standard Form of

PUBLICATIONS

Naval Air Systems Command

ET-590 Avionics Bench, Preflight and Flight Test Instructions for Attitude Gyroscope Control Assembly A/A24G-39

EI-590 Avionic Installation Instructions for Attitude Gyroscope Control Assembly A/A24G-39

Radio Technical Commission for Aeronautics

Document No. DD-160 - Environmental Conditions and Test Procedures for Airborne Electronic/Electrical Equipment and Instruments. (Application for copies should be addressed to the RTCA, Secretariat, Suite 655, 1717 H Street N.W., Washington, D.C. 20006.)

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First Article - When specified in 6.2, the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.2).

3.2 Components - The Attitude Gyroscope Control Assembly shall consist of the following:

a. A/A24G-39 Control Assembly

(1) Gyroscope, Displacement, SBK-11A/A24G-26

(2) Amplifier, Electronic Control,

ASK-32/A24G-39 (fixed wing aircraft)
or, ASK-441A24G (rotary wing aircraft)

(3) Controller, Compass System,

	DCK-128/A24G	(white-lighted)
or,	DCK-129/A24G	(red-lighted)
or,	DCK-228/A24G	(white-lighted, IPL)
or,	DCK-234/A24G	(moisture-sealed, white-lighted)
or,	DCK-235/A24G	(Moisture-sealed, red-lighted)
or,	DCK-236/A24G	(Moisture-sealed, white-lighted, IPL)

3.2.1 Application Considerations - The A/A24G-39 Control Assembly shall be supplied for installations requiring four heading outputs, a single roll output, a single pitch output, and azimuth rate roll erection control. The Compass System Controller is available in different configurations to meet various flight deck lighting requirements and installation considerations (see 6.2). The Electronic Control Amplifier is to be selected to meet the performance criteria of the applicable aircraft (see 6.2).

3.2.2 Associated Equipment - The equipment shall operate properly with the following associated equipment in accordance with 3.7:

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Item	Type Designation	Specification
Magnetic Azimuth Detector	DSU-4A/A	MIL-D-38134
Transmitter, Compass Remote	ML-1	MIL-T-19576

3.3 Selection of standards and Specifications - Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Fungus-proof Materials - Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. If used within a hermetically sealed enclosure, fungicidal treatment shall not be required.

3.4.2 Metals - Metals shall be of the corrosion-resistant type or shall be suitably treated to resist corrosion due to conditions likely to be encountered in storage or normal service.

3.4.2.1 Dissimilar Metals - Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.3 Non-magnetic Materials - Non-magnetic materials shall be used for all parts of the control assembly except where magnetic materials are essential.

3.4.4 Corrosive Fumes - Materials, as installed and used in the control assembly and under the service conditions specified herein, shall not liberate deleterious fumes.

3.4.5 Protective Treatment - When materials are used in the construction of the control assembly that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating which will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5 Design and Construction - The control assembly shall be designed to accept and provide input and output signals as specified herein, and shall be so constructed as to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

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Corresponding components of all control assemblies procured shall be interchangeable between control assemblies- No gimbal binding under differential pressure conditions shall occur as a result of interference between case and gimbal should a leak develop in service.

3.5.1 Plug-in Cards - The electronic control amplifier circuitry and electromechanical modules shall be placed on plug-in cards which shall be removable without the use of special tools. The plug-in cards shall be keyed and color-coded to prevent being installed in the wrong location and shall have test points for easy maintenance-

3.5.2 Maintainability - The following maintainability requirements shall be considered in the design of the control assembly:

a. Minimization of the complexity of maintenance tasks (for example: calibration, adjustments, inspection, et cetera) by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercial items.

b. Rapid and positive recognition of equipment malfunction or marginal performance.

c . Rapid and positive identification of the replaceable defective part, assembly, or components.

d. Minimum numbers and types of tools and test equipment (special and standard) needed to perform maintenance,

e l Optimum accessibility in all systems, equipments, and components requiring maintenance, inspections, removal, or replacement.

3. 5.3 Reliability - The contract specified mean-time-between-failures (MTBF) for the control assemblies and their components shall be as follows based on a discrimination ratio of 2.0 at a confidence level of 80 percent:

a.	A/A24G-39 Control Assembly:	1,350 hours
b.	SBK-11A/A24G-26 Displacement Gyroscope:	2,240 hours
c.	ASK-32/A24G-39 Electronic Control Amplifier:	3,440 hours
d.	ASK-44/A24G Electronic Control Amplifier:	3,440 hours

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e.	DCK-128/A24G Compass System Controller:	10,000 hours
f.	DCK-129/A24G Compass System Controller:	10,000 hours
g.	DCK-228/A24G Compass System Controller:	10,000 hours
h.	DCK-234/A24G Compass System Controller:	10,000 hours
i.	DCK-235/A24G Compass System Controller:	10,000 hours
j.	DCK-236/A24G Compass System Controller:	10,000 hours

3.5.3.1 Longevity - the control assembly and components shall each have an operating life span (equipment longevity as defined in MIL-STD-781) of not less than 2,000 hours before they consistently fail to meet the MTBF index. Those components that are subjected to eliability tests and have individually accumulated 2,000 hours or longer shall not be subjected to a longevity test.

3.6 SBK-11A/A24G-26 Displacement Gyroscope - The displacement gyroscope, including associated electronics, shall consist of a vertical and a directional gyro mounted in multiple gimbals as described in the following paragraphs. This gimbal arrangement is used in order to avoid a gimbal lock position.

3.6.1 Vertical Gyro - The vertical gyro shall have sufficient angular momentum and be so designed as to meet the performance specified herein when operated throughout the temperature, vibration, and acceleration ranges.

3.6.2 Directional Gyro - The directional gyro shall have sufficient angular momentum and be of such design as to provide the performance specified herein when operated throughout the temperature, vibration, and acceleration ranges.

3.6.3 Gimbal System - The displacement gyroscope configuration shall consist of a two-axis directional gyro vertically stabilized by a two-axis vertical gyro.

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3.6.3.1 Inner Directional Gyro Gimbal - The first (inner) directional gyro gimbal (gyro wheel housing) shall be stabilized by the angular momentum of the gyro. Its pivot axis shall be the first directional gyro axis of freedom, shall be perpendicular to the spin axis of the gyro, and normally shall lie in the horizontal plane. The slaving torquer shall act about this axis. The gimbal shall have a minimum of +30° of freedom about its axis.

3.6.3.2 Directional Gyro Azimuth Gimbal - The second directional gyro gimbal shall be stabilized by the angular momentum of the gyro. Its pivot axis shall be the second directional gyro axis of freedom and shall be aligned with the vertical. The azimuth output synchro and the leveling torquer shall act about this axis. The gimbal shall be capable of continuous rotation about its axis.

3.6.3.3 Third Gimbal - The third gimbal shall be the pitch gimbal of the vertical gyro.

3.6.3.3.1 Pitch Gimbal - The first vertical gyro axis of freedom shall be stabilized by the angular momentum of the vertical gyro, shall be normally the pitch axis, and shall normally lie in a horizontal plane. The roll erection torquer shall act about this axis. The gimbal shall incorporate gimbal stops for controlled precession during maneuvers through the vertical. A linear acceleration sensor shall be provided per 3.7.4.2.

3.6.3.4 Fourth (Outer) Gimbal - The fourth gimbal shall be the roll gimbal of the vertical gyro.

3.6.3.4.1 Roll Gimbal - The second vertical gyro axis of freedom shall be stabilized by the angular momentum of the vertical gyro, shall be the roll axis and shall be aligned with the fore-aft axis of the aircraft. The pitch erection torquer shall act about this axis. The gimbal shall have unlimited freedom throughout 360° of roll. A lateral acceleration sensor shall be provided having a threshold acceleration of 0.065g, ±0.01g, directed along the sensitive axis of the acceleration sensor.

3.6.4 Erection - The erection mechanism shall incorporate separate systems for pitch and roll. Erection shall be accomplished by electrolytic switches and torque motors on the gimbals. The erection system shall provide both initial (fast) and final (normal) erection and shall erect the gyro from any attitude to the vertical as determined by the vertical sensing electrolytic switches.

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3.6.4.1 Erection Cycle - The erection cycle shall consist of a period of fast precession erection, not to exceed 120 seconds, after which the control assembly shall revert to the normal erection rate.

3.6.4.2 Initial Erection - Initial erection of the vertical gyro shall be such that it will assume a position within 1° of vertical in pitch and roll within 2 minutes or less after power is applied. The rotor shall be at a standstill and the spin axis may be in any possible attitude at the time power is applied. The vertical gyro shall align its spin axis with the direction of gravity within ±1/4° in 2-1/2 minutes in pitch and in roll.

3.6.4.3 Erection Rates

3.6.4.3.1 Normal Rate - The normal erection rate about the roll axis shall be nominally 1.25° per minute and about the pitch axis shall be 5° per minute for deflections of spin axis greater than 0.2° from vertical. Actual rates shall be as shown on figure 1. Pitch erection sensing shall be maintained when displacing the gyro in roll ±45° or less.

3.6.4.3.2 Airborne Mode of Operation - If desired, the normal erection rate about the pitch axis may be reduced when airborne to 1.25° per minute by grounding connector 2J2-5 of the Electronic Control Amplifier.

3.6.4.3.3 Fast Erection - Means shall be provided to accomplish fast erection of the vertical gyro to the apparent vertical of both axes of the gyro simultaneously. Fast erection excitation shall be made available at connector 1J1-8 of the displacement gyroscope.. The fast erection rate shall be not less than 12° per minute in either axis.

3.6.4.4 Roll and Pitch Pendulum Erection Cutout - The design shall be such that when pins 6 and 7 of table I are connected externally, roll erection will be in effect and when pins 5 and 7 are connected externally, pitch erection will be in effect. When all three pins are connected externally, both roll and pitch erection shall be functioning. The erection mechanism shall be so designed that when pins 6 and 7 of table I are disconnected externally, roll erection will be cut out and when pins 5 and 7 are disconnected externally, pitch erection will be cut out.

3.6.4.5 Manual Fast Erection - Provisions shall be made for manual operation of the fast erection circuits. Actuation of the manual control shall provide fast erection only during actuation and shall not affect the normal erection cycles.

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3.6.4.6 Power Interruption - The control assembly erection system shall not revert to the fast precession erection rate after a power interruption of less than 25 seconds.

3.6.5 Vertical Gyro Synchro Outputs - One transmitting synchro shall be mounted on each vertical gyro axis of freedom. The synchros shall provide displacement signals of gimbal movement about their respective axes at a 1 to 1 ratio to the amplifier. The nominal electrical characteristics shall be in accordance with table II. The synchros shall be internally excited from øA (see table I).

3.6.5.1 Pitch Synchro - When pin 11 of table I is connected to pin 21, and the gimbals positioned for zero pitch displacement, the voltage across pins 10 and 12 shall be minimum and the voltage from pin 10 to 22 shall be greater than the voltage from pin 22 to pin 21. When the gimbals are rotated in a nose-up direction, the voltage from pin 10 to pin 11 shall increase before decreasing.

3.6.5.2 Roll Synchro - When pin 14 of table I is connected to pin 21, and the gimbals positioned for zero roll displacement, the voltage across pins 13 and 15 shall be minimum and the voltage from pin 13 to pin 22 shall be greater than the voltage from pin 22 to pin 21. When the gimbals are rotated in a right-wing-down direction, the voltage from pin 13 to pin 14 shall increase before decreasing.

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TABLE 1. SBK-11A/A24G-26 Displacement Gyroscope Connector to Electronic Control Amplifier

Pins 1/	Function
1 (S1)	Heading output, øA excited
2 (S2)	Heading output, øA excited
3 (S3)	Heading output, øA excited
4	Erection warning
5	Pitch erection
6	Roll erection
7	Test Connection, 26V øC
8	Fast erection excitation, 78V øC
9	Lateral acceleration sensor
10 (S1)	Pitch output, øA excited
11 (S2)	Pitch output, øA excited
12 (S3)	Pitch output, øA excited
13 (S1)	Pitch output, øA excited
14 (S2)	Pitch output, øA excited
15 (S3)	Pitch output, øA excited
16	Linear acceleration sensor
17	N. Latitude control and slaving
18	S. Latitude control and slaving
19	S pare
20	Spare
21'	Ground - neutral
22	115V øA
23	115V øB
24	115V øC

1/ Connector (1J1) MS24264R16B24PNX to mate with MS24266R16B24SN

3.6.6 Vertical Drift - The vertical gyro assembly shall be so designed that the maximum free drift, noted after compensation for the earth's rotation, shall be not greater than 0.200 per minute under standard conditions and 0.25° per minute over the temperature range specified in 3.9.

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3.6.7 Directional Gyro Leveling - The inner gimbal shall be precessed by means of the leveling torquer controlled by a suitable non-pendulous sensor so that the spin axis of the directional gyro is perpendicular to the azimuth axis within ±1/4°. This leveling torquer shall have a rate that will level the gyro—within 30 seconds after power is applied.

TABLE II. Displacement Gyroscope Synchros
Nominal Electrical Characteristics

Internal excitation	26V øA, 400 Hz
Input current	80 mA
Input power	0.44w
Output voltage (max)	11.8V
Sensitivity (no load)	206 mV per deg
Phase shift (no load)	7.50
dc rotor resistance	43 ohms
dc stator resistance	30 ohms
Impedance	Z_{ro} = 69 + j318 ohmS
Impedance	Z_{so} = 28 + j55 ohms
Impedance	Z_{rss} = 142 + j53 ohmS
Maximum null	50 mV
Accuracy (max error spread)	14 minutes

3.6.8 Directional Gyro Slaving - The directional gyro shall be precessed about its azimuth axis to an external azimuth signal by means of the azimuth torquer. The normal slaving rate shall be 10 to 20 per minute.

3.6.9 Azimuth Transmitting Synchro - One transmitting synchro shall be mounted on the azimuth axis of freedom. The synchro shall provide displacement signals of gimbal movement about the heading axis at a 1 to 1 ratio to the amplifier. The nominal electrical characteristics shall be in accordance with table II. The synchro shall be internally excited from øA (see table I).

3.6.9.1 Synchro Rotation - The gimbals shall be positioned to obtain a minimum voltage across pins 1 and 3 of table I, and then rotated in a right-turn direction. The voltage from pin 1 to pin 2 shall increase before decreasing.

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3.6.10 Balance - The directional gyro shall be so balanced that the apparent drift due to the earth's rotation is equal to the product of the sine of the latitude of operation and 15° per hour in a counterclockwise (ccw) direction in the Northern Hemisphere and in a clockwise (cw) direction in the Southern Hemisphere.

3.6.11 Maximum Drift - The maximum drift of the directional gyro shall not exceed ±1.00 per hour from the computed apparent drift for the latitude of operation at room temperature.

3.6.11.1 Random Drift - With normal distribution, the random drift of the directional gyro shall not exceed ±0.250 per hour - one sigma.

3.6.12 Electrical Connector - The electrical connector shall be in accordance with table I and figure 2.

3.6.13 Weight - The weight of the displacement gyroscope shall not exceed 10.2 pounds including the vibration isolator.

3.6.14 Dimensions - The dimensions of the displacement gyroscope shall be in "accordance with figure 2.

3.6.15 Mounting - The mounting of the displacement gyroscope shall be in accordance with figure 2 and shall be such that the displacement reference will not deflect by more than ±0.1° in pitch, roll, or azimuth as a result of accelerations or any vibration between 0 and 18 Hz with a double amplitude of 0.020 inch. The torsional resonant frequency shall be greater than 18 Hz in pitch, roll, and azimuth.

3.6.16 Elapsed Time Indicator - The displacement gyroscope shall contain an elapsed time indicator (ETI) conforming to MS17322 and MIL-M-7793 having a range of at least 9,999 hours. The ETI shall be located as shown in figure 2 and shall be visible without disassembly of the displacement gyroscope case.

3.6.17 Color - The color of the displacement gyroscope shall be gray in accordance with FED-STD-595, Color No. 36231.

3.7 ASK-32/A24G-39 or ASK-44/A24G Electronic Control Amplifier - The electronic control amplifier shall contain the azimuth servo amplifier, slaving amplifier, deviation compensator, turn and acceleration control functions, fast synchronization circuitry, malfunction monitoring circuitry, and associated power supplies. It also shall function as a system interconnection box with connectors as shown in tables III, IV, V,

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and VI. Provisions shall be made for the excitation of the magnetic azimuth detector, DSU-4A/A, conforming to MIL-D-38134, or the ML-1 connforming to MIL-T-19576.

TABLE 111. ASK-32/A24G-39 or ASK-44/A24G Electronic Control Amplifier Connector to Displacement Gyroscope

Pins 1/	Function
1 (S1)	Azimuth input
2 (S2)	Azimuth input
3 (S3)	Azimuth input
4	Erection warning
5	Linear acceleration signal
6	Not used
7	Pitch control
8	Roll control
9	Ground (neutral)
10	115V øA
11	115V øB
12	115V øC
13	Fast erection control excitation, 78V øC
14	Spare
15	Spare
16	Spare
17	N. latitude control and slating
18	S. latitude control and slating
19 (S1)	Pitch input
20 (S2)	Pitch input
21 (S3)	Pitch input
22 (S1)	Roll input
23 (S2)	Roll input
24 (S3)	Roll input

1/ Connector (2J2) MS24264R16B24S6X to mate with MS24266R16B24P6

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TABLE IV. ASK-32/A24G-39 or ASK-44/A24G Electronic Control Amplifier Connector to Compass System Controller and Magnetic Azimuth Detector

Pins 1/	Function
1	Heading set relay
2	Heading set input
3	26V øA
4	Latitude control excitation +27V
5	Mode selector (SLAVED)
6	Sync indicator (+)
7	Sync indicator (-)
8	Latitude correction - South
9	Latitude correction - North
10	Slaving common (ground)
11	Malfunction signal 26V øA
12	Fast erection relay
13	Automatic pilot cutoff normally open
14	Automatic pilot cutoff normally closed
15	Automatic pilot cutoff common
16 (S1)	Magnetic Azimuth detector input
17 (S2)	Magnetic Azimuth detector input
18 (S3)	Magnetic Azimuth detector input
19	Shield
20 (R1)	Magnetic Azimuth detector excitation
21 (R2)	Magnetic Azimuth detector ground
22	Malfunction signal (-28Vdc)
23	Controller ground
24	Mode selector (COMP or EMERG) 2/

1/ Connector (2J3) MS24264R16B24SNX to mate with MS24266U6B24PN

2/ Mode Selector nomenclature is assigned by type designation of the Compass System Controller as noted in figures 4 and 5.

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TABLE V. ASK-32/A24G-39 or ASK-44/A24G Electronic Control Amplifier Connector - Power Inputs and Heading Outputs

Pins 1/	Function	
1 (S1)	Heading output No. 1	Primary
2 (S2)	Heading output No. 1	
3 (S3)	Heading output No. 1	
4 (S1)	Heading output No. 2	
5 (S2)	Heading output No. 2	
6 (S3)	Heading output No. 2	
7 (S1)	Heading output No. 3	
8 (S2)	Heading output No. 3	
9 (S3)	Heading output No. 3	
10 (S1)	Heading output No. 4	Secondary
11 (S2)	Heading output No. 4	
12 (S3)	Heading output No. 4	
13 (R1)	Heading excitation No. 1	
14 (R2)	Heading excitation No. 1	
15 (R1)	Heading excitation No. 2	
16 (R2)	Heading excitation No. 2	
17 (R1)	Heading excitation No. 3	
18 (R2)	Heading excitation No. 3	
19 (R1)	Heading excitation No. 4	
20 (R2)	Heading excitation No. 4	
21	Spare	
22	Malfunction signal (-28 Vdc)	
23	Ground - Neutral	
24	115V øA	Input Power
25	115V øB	
26	115V øC	
27	Malfunction relay (common)	
28	Spare	
29	Spare	
30	Malfunction relay (N.C.)	
31	Spare	

1/ Connector (2J1) MS24264R18B31PNX to mate with MS24266R18B31SN

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TABLE VI. ASK-32/A24G-39 or ASK-44/A24G - Electronic Control Amplifier Connector - Pitch and Roll Outputs

Pins 1/	Function
1 (S1)	Pitch output
2 (S2)	Pitch output
3 (S3)	Pitch output
4 (S1)	Roll output
5 (S2)	Roll output
6 (S3)	Roll output
7	Spare
8	Spare
9	Spare
10	Spare
11	Spare
12	Spare
13	External fast erection control
14	Spare
15	Spare

1/ Connector (2J4) MS24264R14B15SNX to mate with MS24266R14B15PN

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3.7.1 Amplifier Channels - The amplifier channels shall contain modular plug-in assemblies (3.5.1).

3.7.2 Fast synchronization and Heading Set Drive - An electromechanical mechanism shall be provided to cause synchronization of the azimuth servo to the magnetic azimuth detector when operating in the SLAVED mode and shall position the azimuth servo to any desired reference to set the heading outputs when operating the control assembly in the DG mode. The mechanism shall have a drive rate continuously variable in either direction up to 30° per second as selected by the heading set control specified in 3.8.3. The mechanism shall be suitably damped to permit accurate synchronization and selection of the desired reference.

3.7.2.1 Push-to-Synchronize Circuit - The design of the circuit shall be such that when the heading set control (3.8.3.1) is pressed and held, and the mode selector (3.8.2) is in the SLAVED mode of operation, the control assembly shall synchronize to the magnetic azimuth detector heading:

a. within 6 seconds for the ASK-32/A24G-39 electronic control amplifier.

b. within 8 seconds for the ASK-44/A24G electronic control amplifier.

3.7.2.2 Automatic Autopilot Decoupling - The continuity between connector 2J3-14 and 2J3-15 of table IV shall be transferred to between connector 2J3-13 and 2J3-15 incident to and during sychronization, slewing, and compass mode conditions only. The design shall be such that the transfer must be made before fast synchronization can be actuated.

3.7.3 Heading Servos - the heading servo shall have a minimum follow-up rate of 60° per second to maintain synchronization with the displacement gyroscope signals throughout all maneuvers. Operation shall be smooth with no lag in excess of 0.3° per rpm up to 5 rpm in heading. No oscillation, hunting, or jumping of a magnitude or frequency that will affect the operation of a high-performance automatic flight control system shall occur. The undamped natural frequency shall be greater than 6 Hz. The heading servo shall position the rotors of four data transmitter synchros.

3.7.4 Magnetic slaving and and Erection Control - Provisions shall be made for erection control, and where applicable, for interrupting slaving of the directional gyro. Erection control shall operate under all modes of operation. Magnetic slaving cutout shall be effective when the control assembly is in the SLAVED mode. Actuation circuitry for turn rate control and fore-aft acceleration control shall be contained in the electronic control amplifier.

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3. 7.4.1 Roll Erection and Magnetic Slaving Control

3. 7.4.1.1 ASK-32/A24G-39 Amplifier - An azimuth rate sensor shall be provided to permit control of roll erection and magnetic slaving during turns. Under standard conditions, roll erection and magnetic slating control shall occur for the following azimuth inputs:

a. Roll erection and magnetic slaving shall be removed (OFF) during turns of ±60, ±30°, and ±180° per minute within 30, 11, and 8 seconds respectively, and shall be returned (ON) within 18, 26, and 35 seconds, respectively, after the rates have ceased. Erection and slaving shall remain ON during turns of ±3° per minute or less.

b. Roll erection and magnetic slaving shall be removed (OFF) within 30 seconds when an input of 30° per minute rate of turn is superimposed on a ±0.8°, 1-second period azimuth oscillation. Erection and slaving shall remain OFF until the 30° per minute rate of turn has ceased.

c. Roll erection and magnetic slaving shall remain ON when the following azimuth oscillations are imposed:

(1) ±0.8° with a 1.0 second period.

(2) ±2.0° with an 8.0 second period.

3.7.4.1.2 ASK-44/A24G Amplifier - An azimuth rate sensor shall be provided to permit control of roll erection and magnetic slaving during turns. Under standard conditions, erection and slaving control shall occur for the following azimuth inputs:

a. Roll erection and magnetic slaving shall be removed (OFF) during turns of ±24° and +±80° per minute within 12 and 4 seconds, respectively, and shall be returned (ON) within 8 and 14 seconds, respectively, after the rates have ceased. Erection and slaving shall remain ON during turns of ±12° per minute or less.

b. Roll erection and magnetic slaving shall remain ON when an azimuth oscillation of ±0.80 with 1.0 second period is imposed.

3.7.4.2 Linear Acceleration Control - An internal sensor on the displacement gyroscope and the associated circuitry shall be provided to permit control of pitch erection and (as applicable) magnetic slaving during horizontal fore-aft acceleration. Control shall occur at a threshold acceleration of 0.065g, ±0.01g, directed along the sensitive axis of the acceleration sensor. During longitudinal accelerations in excess of 0.066g, ±0.01g, the following shall occur:

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a. The pitch erection shall be reduced to 25 percent of the nominal value and magnetic slaving shall be removed when using the ASK-32/A24G-39 electronic control amplifier.

b. The pitch erection shall be reduced to 25 percent of the nominal value when using the ASK-44/A24G electronic control amplifier.

3.7.5 Data Transmitters - Four synchro transmitters whose rotors are positioned by the heading servo shall be provided to transmit information. Each synchro transmitter shall have nominal electrical characteristics as shown in table VII. Three of these transmitters shall be designed primary and identified as No.'s 1, 2, and 30 The fourth shall be designated secondary (table V).

TABLE VII. Heading Output Transmitters Synchro Nominal Electrical Characteristics

Input voltage	115V 400 Hz
Input current	29 mA
Input power	0.8 W
Output voltage (max)	11.8V
Sensitivity (no load)	206 mV/deg
Phase shift (no load)	110
dc rotor resistance	700 ohms
dc stator resistance	10.4 ohms
Impedance	Z_{ro} = 950 + j3,850 ohms
Impedance	Z_{so} = 10 + j36 ohms
Impedance	Z_{rss} = 1,550 + j420 ohms
Max null voltage	75 mV
Accuracy (max error spread)	14 minutes

3.7.5.1 Data Transmitter Zero and Sensing - Input transmitters for simulating heading, pitch, and roll signals from the displacement gyroscope shall be indexed and coded for electrical zero and rotation in accordance with MIL-S-20708. These transmitters shall be excited from 400 Hz øA, with R1 high and R2 low. Stator connections to the amplifier shall be in accordance with table III.

3.7.5.1.1 Heading Transmitter - Rotor and stator leads of each heading output transmitter are identified in table V in accordance with the lead coding standards of MIL-S-20708. With R2 of each output transmitter connected to S2, and with R1 to R2 excited from 400 Hz, any phase, turn the heading input transmitter to obtain a minimum voltage from S1 to S3 and a voltage greater than the excitation voltage from S1 to

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R1. Positive rotation of the input transmitter (to simulate a right turn) shall cause the stator voltage of each output transmitter to become maximum in the sequence S2-1, S1-3, S3-2.

3.7.5.1.2 Pitch and Roll Transmitters - Stator leads of the pitch and roll output transmitters of the displacement gyroscope shall be routed through the electronic control amplifier, and shall be available at connector 2J4. The connections are identified in table VI in accordance with the lead coding standards of MIL-S-20708.

3.7.6 Malfunctioning Monitoring - Provisions shall be made for detecting the following faults:

a. Supply voltage high (indication initiated at between +10 and +20 percent).

b. Supply voltage low (indication initiated at between -10 and -20 percent).

c. Loss of synchro excitation caused by an open in the power source or in the rotor of the gyroscope heading synchro transmitter.

d. Loss of synchro signal transmission caused by an open in any two stator connections from the gyroscope heading synchro transmitter including opens in the aircraft interconnecting wiring between the gyroscope and the electronic control amplifier.

e. Heading servo followoo-up displaced from null. (Indication initiated at between 2° and 3° error signal).

3.7.6.1 Malfunction Indication

a. A negative 28Vdc signal shall be removed from connector 2J3-22 and 2J1-22 when any fault in the heading channel is detected as specified in 3.7.6.

b. A set of normally open contacts shall be provided between connector 2J1-27 and 2J1-30. These shall close when any fault is detected as specified in 3.7.6.

3.7.6.2 Auxiliary Malfunction indication - A 26V øA signal shall be provided at connector 2J3-11 when any fault detected as specified in 3.7.6.

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3.7.6.3 Fast Erection Indication - In the SLAVED and DG modes, the malfunction output signals shall be present until completion of the erection cycle and during activation of fast erection.

3.7.6.4 Negative 28Vdc Malfunction Signals - The negative 28Vdc signals provided at connector 2J1-22 and 2J3-22 shall have a combined load capability of 150 milliamps.

3.7.7 Deviation Compensation - A 24-point electrical heading deviation compensator shall be provided having a nominal adjustment range of ±4° and a repeatable accuracy of ±9 minutes at each 15° compensation point. The deviation compensation-assembly shall be modular and plug-in, allowing transfer to another electronic control amplifier without major disassembly of either amplifier and without loss of the stored compensation information.

3.7.8 Dimensions - The dimensions of the electronic control amplifier shall be in accordance with figure 3.

3.7.9 Weight - The weight of the amplifier shall not exceed 7.10 pounds.

3.7.10 Elapsed Time Indicator - The electronic control amplifier shall contain an elapsed time indicator (ETI) conforming to MS17322 and MIL-M-7793 having a range of at least 9,999 hours. It shall be visible without any disassembly of the amplifier per figure 3.

3.7.11 Color - The color of the electronic control amplifier shall be gray in accordance with FED-STD-595, Color No. 36231.

3.8 Compass System Controllers - The compass system controller shall be designed in accordance with MIL-C-6781, Type 1. The configuration shall be in accordance with either figure 4 or 5. Fasteners shall conform to MIL-F-25173 and plastic plate lighting shall conform to MIL-P-7788, Type IV except as follows:

a. Paragraph entitled “Connector Location” shall not apply.

b. The contrast, stray light, and dial visibility requirements of MIL-L-25467 shall apply. Stray light shall be measured 1 inch in front of the panel.

c. the DCK-128/A24G and DCK-234/A24G lighting (white, 4.5V) shall meet the color requirements of MIL-L-27160 and the brightness requirements of MIL-P-7788.

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d. The DCK-129/A24G and DCK-235/A24G lighting (red, 5.0V) shall meet the color requirements of MIL-L-25467 and the brightness requirements of MIL-P-7788.

e. The DCK-228/A24G and DCK-236/A24G lighting (white 5.0V) shall meet the color requirements for Instrument and Panel Lighting, Type 1 (g) of MIL-C-25050 and the brightness requirements of MIL-P-7788.

f. The lighting of the mode selector switch knob for the DCK-234/A24G, DCK-235/A24G, and DCK-236/A24G shall conform to the requirements of MS21384-16.

3.8.1 Prime Standard - The prime standard shall meet the requirements of 3.8. Panel lighting shall be compared to and shall balance in brightness and color with the prime standard.

3.8.2 Mode Selector Switch Functions - A mode selector switch shall be provided for selecting any of the following three modes of operation:

a. SLAVED - In the SLAVED mode, the control assembly shall operate as a gyro stabilized magnetic compass.

b. DG - In the DG mode, the control assembly shall provide heading when operated independently of the magnetic azimuth detector as a latitude corrected directional gyroscope.

c. COMP (compass) or EMERG (emergency) - In the compass or emergency mode, the control assembly shall be directly slaved to the magnetic azimuth detector. During this mode, the autopilot interlocks shall be activated. Use of this mode shall be limited to flight emergencies only.

3.8.2.1 Mode Selector Knob - Each of the compass system controllers shall have a mode selector switch knob which shall conform to the requirements of MIL-K-25049 in accordance with the following:

MS25169P2 with Ms5164 Type I Skirt	MS21384-16 Illuminated
DCK-128/A24G	DCK-234/A24G
DCK-129/A24G	DCK-235/A24G
DCK-228/A24G	DCK-236/A24G

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3.8.3 Synchronization and Heading Set - A manual control as shown in figures 4 and 5 shall be provided to effect the fast synchronization and heading set as specified in 3.7.2. When in the DG mode, the heading set control markings shall indicate in which direction the control knob is to be turned to provide a decreasing or increasing heading. The control shall be of the push-to-turn, spring return-to-center type.

3.8.3.1 Push-to-Sync - When the heading set control is pushed, the push-to-synchronize circuit specified in 3.7.2.1 shall be actuated.

3.8.3.2 Timed Automatic Synchronization - provisions shall be included in the following controllers to initiate a 10 ±2 second automatic fast synchronization cycle whenever the mode selector switch is changed from either EMERG or DG to the SLAVED mode.

DCK-234/A24G
DCK-235/A24G
DCK-236/A24G

3.8.3.3 Fast Erection - The compass system controller shall incorporate means to operate the fast erection relay in the electronic control amplifier to accomplish fast erection of the vertical gyro specified in 3.6.4.3.3.

a. For the following compass system controllers, pushing the heading set control knob shall cause fast erection if the aircraft lead between connector 3J1-12 and connector 2J3-12 is installed as shown in figure 6.

DCK-128/A24G
DCK-129/A24G
DcK-228/A24G

(1) An alternative means for fast erection when utilizing any of the above controllers if the wire between connector 3J1-12 and connector 2J3-12 is not installed, is to provide an external normally open spring-return switch connected between connector 2J4-13 and power ground.

b. For the following compass system controllers a separate switch labeled ERECT, when pushed, shall cause fast erection. An aircraft interconnection must be provided between connector 3J1-12 and connector 2J3-12 as shown in figure 6.

DCX-234/A24G
DCK-235/A24G
DCK-236/A24G

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3.8.4 Latitude Correction Control - A manual set control labeled LAT with a latitude dial shall be provided to correct the control assembly for the apparent drift of the directional gyro due to the earth's rotation. The latitude dial shall be marked in latitude from 0° to 90, graduated in 2° increments, with major divisions and numerals 10°. The correction shall be applied in both DG and SLAVED modes of operation.

3.8.4.1 North and South Hemisphere Selector - The selection of the Northern or Sourthern Hemisphere shall be made by a manual selector and the selection shall be indicated by the display of the letter on the compass system controller panel.

3.8.5 Synchronizing Indicator - A synchronizing indicator, other than signal lights, shall be provided to indicate synchronization to within ±1/40 between the heading output and the magnetic azimuth detector, or electrical equivalent, when in the SLAVED mode of operation. The synchronizing indicator lighting and markings shall afford ease of viewing under all ambient lighting conditions. The synchronizing indicator shall remain fixed in the centered position during DG mode of operation

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TABLE VIII. DCK-(*)/A24G Compass System Controller Connector to Electronic Control Amplifier

Pins 1/	Function
1	Heading set control
2	Heading set signal output
3	26V øA
4	+27V
5	Mode select (SLAVED)
6	Sync indicator
7	Sync indicator
8	Latitude correction South
9	Latitude correction North
10	Slaving common
11	Spare (NO NOT USE)
12	Fast erection control
13	Spare
14	Spare
15	Spare
16	Spare
17	Spare
18	Spare
19	Spare
20	Spare (DO NOT USE)
21	Panel lighting (5V)
22	Panel lighting (common)
23	Ground
24	Mode select (COMP or EMERG)

1/ Connector (3J1) MS24264R16B24P6X to mate with MS24266R16B24S6

*Includes all Compass System Controllers referenced in Section 3.2.a(3).

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3.8.6 Markings and Dimensions

3.8.6.1 Marking - All markings shall be so durable as to withstand usage encountered in service. Unless otherwise specified, the style and proportions of numerals and letters shall conform to MS33558.

3.8.6.2 Numerals - Each numeral shall be so distinctly marked that no doubt will exist as to which graduation the numeral applies.

3.8.6.3 Marking Dimensions - The dimensions of the markings shall be as shown in table IX.

TABLE IX. Compass System Controller Marking and Dimensions

	Height or Length Inch ±0.005	Width of Line or Graduation Inch ±0.005
Letters or Words	0.14	0.020
Numerals	0.12	0.020
2° graduations	0.10	0.015
10° graduations	0.14	0.020
"+" and "-"	0.12	0.020

3.8.6.4 Color - The background of the dial and all other markings not otherwise specified shall be finished in durable dull black in accordance with FED-STD-595, Color No. 37038. Knobs shall be gray, Color No. 36231 of FED-STD-595 and all markings shall be white, color No. 37875 of FED-STD-595.

3.8.7 Weight - The weight of the compass system controller shall not exceed 1.35 pounds, except for the compass system controllers with the timed automatic synchronization circuit specified in 3.8.3.2, wich shall not exceed 1.5 pounds.

3.8.8 Connector - The connector shall be in accordance with table VIII.

3.9 Performance - The control assembly and each component shall be capable of meeting the tests specified in section 4 under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for class 2 equipment except as modified herein:

a. Operating temperatures ranging from -54° to +71°C (4.6.12).

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b. Extended storage at -65°C and 10 minutes at +98°C (except for the compass system controller where the upper temperature limit is +85°C).

c. Dielectric strength of 375Vac or 500Vdc.

d. Vibration as specified in 4.6.8 and 4.6.19.

e. Pressure equivalent to 70,000 foot altitude (4.6.12).

f. Shock forces of 15 g's except for crash safety which shall be 40 g's (4.6.22).

g. Humidity up to and including 95 percent (4.6.13).

h. Exposure to fungus enzymes (4.6.14).

i. Exposure to a simulated rainfall (4.6.15).

j. Salt spray for a period of 48 hours (4.6.16).

k. Exposure to blowing fine sand and dust particles (4.6.17).

l. Operation in explosive vapors (4.6.21).

m. Exposure to sudden temperature changes (4.6.18).

3.10 Radio Noise Interference and Susceptibility. The control assembly and each component shall meet the radio noise interference and susceptibility requirements of MIL-I-6181.

3.11 Interchangeability - All pares having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-1000 shall govern the manufacturer's part number and changes thereto.

3.12 Electronic Components - The electronic components shall be in accordance with MIL-E-5400 where applicable.

3.13 Electrical Power Requirements - The power source required for the control assembly shall be 4-wire, 115/200V, 3-phase, 400 Hz, wye (nominal) in accordance with MIL-STD-704, category B and as specified herein. The compass system controller lighting shall utilize 5V ac or dc at 1.0A maximum.

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3.13.1 Electrical Power Variation - The control assembly shall operate within specified performance limits from a category B power source as defined in MIL-STD-704.

3.13.2 Phase Rotation - The phase rotation of the 400 Hz power required by the control assembly shall be A-B-C with neutral as ground.

3.14 Electrical Connectors - Unless otherwise specified, electrical connectors shall be in accordance with MIL-C-26500.

3.15 Test Pins - Adequate test pins shall be provided on each component for external testing of the item.

3.16 Bonding - Adequate electrical bonding points shall be provided on each component of the attitude gyroscope control assembly.

3.17 Cases - The cases shall be in accordance with the specified dimension shown in figures 2, 3, 4, and 5 and shall provide a hermetically sealed enclosure when required for the internal mechanism. The design of the case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed without the use of special tools and fixtures. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials that will be affected by the action of any atmosphere to which the control assembly may be subjected and in such a manner that it will meet the tests specified in section 4.

3.17.1 Hermetic Sealing - The following hermetic sealing requirements shall apply except in those extreme cases where the filling medium or the pressure, or both, materially affects the performance of the instrument. Exceptions shall be based upon sound engineering facts and shall be presented to the procuring activity for approval. Hermetic sealing is defined in 6.3.1.

3.17.1.2 Filling Medium - The filling medium shall be a mixture of 98 percent helium and 2 percent nitrogen or 95 percent helium and 5 percent carbon dioxide and shall contain a maximum of 0.006 milligram of water vapor per litre (dewPoint 65°C) at the filling pressure. The absolute pressure of the filling medium shall be 12 to 14.7 psi.

3.18 Soldering - Soldering shall be accomplished in accordance with MIL-STD-454, requirement 5.

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3.19 Screw Threads - Unless otherwise specified herein, the threads of all machine screws shall conform to MIL-S-7742.

3.20 Weight - The weight of the control assembly shall not exceed 18.56 pounds, or 18.71 pounds with the compass system controller specified in 3.8.7.

3.21 Finishes and Protective Coatings

3.21.1 Aluminum Alloy Parts - Aluminum parts shall be protected according to MIL-E-5400.

a. Dials, small holes, and case inserts need not be anodized.

b. When abrasion resistance is a factor, chemical film in accordance with MIL-C-S541 shall not be used in lieu of anodizing.

3.21.2 Ferrous Alloys - Ferrous materials shall be protected in accordance with MIL-STD-454, Requirement 15.

3.22 Identification of Product - Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-N-18307.

3.23 Workmanship - The control assembly, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, brazing, painting, riveting, machine screw assemblies, and freedom of parts from burrs and sharp edges.

3.23.1 Dimensions and Tolerances - Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the system, they shall be held or limited accordingly.

3.23.2 Screw Assemblies - Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt threads.

3.23.3 Riveting - Riveting operations shall be performed carefully to insure that the rivets are tight and satisfactorily headed.

3.23.4 Gears - Gear assemblies shall be properly aligned and meshed and shall operate without interference, tight spots, loose spots or other irregularities. Where required for accuracy adjustment, gear assemblies shall be free from backlash.

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3.23.5 Cleaning - The control assembly components shall be cleaned thoroughly of loose, spattered or excess solder, metal chips, or other foreign material after assembly. Burrs and sharp edges as well as resin flash which might crumble shall be removed.

3.24 Lubrication - The displacement gyroscope rotor and gimbal, and electronic control amplifier electro-mechanical assembly bearings shall be lubricated.

3.24.1 Rotor Bearing Lubrication - The rotor bearings shall be lubricated with grease conforming to MIL-G-81322 or MINIPURE grease with particle size not exceeding 50 microns.

3.24.2 Bearing Lubrication - The lubricating oil used in gimbal bearings shall conform to the requirements of MIL-L-81846.

3.24.3 Barrier Film Bearing Coating - All bearings shall be processed in accordance with MIL-STD-1334. The coating material shall be in accordance with MIL-B-81744. All bearings, including chamfered races or corners, and including interfacing parts where applicable, shall be barrier coated.

3.24.4 Ball Bearings - Ball beatings shall be ABEC-5 grade or better. Teflon separators shall be used on gimbal bearings where the bearing design permits. Spin rotor bearings shall have porous separators impregnated with lubricant MIL-L-81846 and shall be ABEC-7 or better.

3.24.5 Ball Bearing Pre-Loading - Means shall be provided for preloading and alignment without causing eccentric loading unless otherwise authorized.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Classification of Inspection - The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.4).

b. Quality Conformance Inspection (see 4.5).

4.3 Test Conditions

4.3.1 Standard Conditions - Unless otherwise specified, the standard test conditions shown in table X shall be used:

TABLE X. Standard Test Conditions

Temperature	Room temperature (25° ±10°C)
Altitude	Normal Ground
Vibration	None
Humidity	Room ambient up to 95 percent relative humidity
Attitude	Level
Input Voltages	a. Three phase, 400 +4 Hz, wye, 115 ±1V to common neutral, phase rotation A, B, C b. 5 ±0.1V ac or dc for lighting

4.3.2 Special Test Equipment - The following special test equipment shall be used:

4.3.2.1 Scorsby Table - The scorsby table shall make an excursion about each axis of ±1 1/2° at 5 to 7 cycles per minute with motion reversal once each minute.

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4.3.2.2 Indexing Fixture - The indexing fixture shall be so designed that it can be indexed to ±90° in pitch and 360° in roll from a horizontal attitude and can be accurately positioned through 360° in heading.

4.3.2.3 Servoed Repeater - The servoed repeater shall be scale graduated in 2 minute angular intervals or smaller and shall have a resolution of 1 minute or less.

4.3.2.3.1 Lead Coding - The servoed repeater shall be so connected as to excite the rotor and to receive the stator signals of a synchro transmitter that has been previously adjusted and coded for electrical zero and rotation per MIL-S-20708. These connections shall be arranged to obtain a zero angular readout with the transmitter on electrical zero, and an increasing angular readout as the transmitter stator voltages become maximum in the sequence S2-1, S1-3, S3-2. The servoed repeater leads shall be coded R1, R2, S1, S2, S3 to correspond with the coding of the associated transmitter leads.

4.3.2.4 Simulated Gyroscope Transmitter - The simulated gyroscope transmitter shall be a Clifton synchro transmitter, type CGH-11-B-2 mounted in an indexing head, or equivalent, and adjusted and coded for electrical zero and rotation in accordance with MIL-S-20708. The rotor shall be excited from 26 ±1V øA with R1 high and R2 low.

4.3.2.5 Simulated Azimuth Detector - The simulated magnetic azimuth detector shall be a Clifton synchro transmitter, type CGH-11-B-2, or equivalent, mounted in an indexing head or equivalent and adjusted and coded for electrical zero and rotation in accordance with MIL-S-20708. The rotor shall be excited from 230 ±10 mV 800 Hz with R1 high and R2 low The high side of the 800-Hz excitation voltage is identified with the high side of the 400 Hz, ±A reference voltage as follows:

a. The 800 Hz excitation voltage passes through zero amplitude at the same instant that the 400 Hz reference voltage passes through zero amplitude.

b. The 800 Hz excitation voltage increases negatively from zero amplitude at the same instant that the 400 Hz reference voltage increases positively from zero amplitude.

4.3.2.6 Simulated Compass Controller - The simulated compass controller provides controls and signals simulating those functions of the compass system controller required for conducting tests of the electronic control amplifier. An actual compass system controller as shown in figure 4 or 5 will be acceptable.

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4.3.2.7 Motor-drive Simulated Gyroscope Transmitter

4.3.2.7.1 The motor-driven simulated gyroscope transmitter shall be similar to that specified in 4.3.2.4 except it shall be so designed that it can be driven at 5 and 50 rpm ±10 percent and, in addition, shall be driven as follows:

a. Steady-state: 60° per second ±10 percent; 180° per minute ±10 percent; 30° per minute ±10 percent; 24° per minute ±3 percent; 12° per minute 13 percent; 6° per minute ±3 percent, and 3° per minute ±3 percent. The 3°, 6°, 12°, 24°, 30°, and 1800 per minute rat—es shall be reversible.

b. Oscillatory (sinusoidal): ±2.0° with an 8 second period and ±0.80 with a 1 second period.

c. Superimposed steady-state and oscillatory: 300 per minute superimposed on a ±0.8°, 1 second period oscillation.

4.4 First Article Inspection - First article consists of examinations and tests performed on samples specified in 4.6 (which are representative of the production item) after the award of a contract. The examinations and tests are conducted to determine that the control assemblies meet the requirements of this specification.

4.4.1 First Article Samples - The test samples shall consist of the equipment and quantities shown in table XI. Three samples of each control assembly (or component, as applicable) shall be subjected to all the tests except reliability and the remaining samples shall be subjected to the reliability test only. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

TABLE XI. First Article Test Samples

Equipment	No. of Samples
A/A24G-39 control assembly	12
SBK-11A/A24G-26 displacement gyroscope	15
ASK-32/A24G-39 electronic control amplifier	13
AsK-44/A24G electronic control amplifier	13
DCK-128/A24G, DCK-129/A24G, or DCK-228/A24G compass system controller	15
DCK-234/A24G, DCK-235/A24G, or DCK-236/A24G compass system controller	15

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4.4.1.1 Reliability Demonstration - Unless otherwise specified by the procuring activity (see 6.2), the control assemblies or their components (see 4.4.1) shall be subjected to the reliability tests specified under 4.6.24, as applicable. Reference MIL-STD-781 section entitled "'Qualification Phase of Production Reliability Tests." The Accept-Reject Criteria for Test Plan III shall be used (3.5.3). Test level F shall be used, except that the temperature extremes shall be +55°C and -44°C, and vibration shall be 0.5g. In the event of a reject decision, the samples will be considered unacceptable and the testing shall be stopped immediately until further notice from the procuring activity.

4.4.2 First Article Test Report - A First Article Test Report shall be prepared in accordance with MIL-STD-831 and two complete copies of the report shall be made available to the procuring activity. Prior to performing the First Article testing, the contractor shall obtain procurement activity approval of the test plan (see 3.1).

4.5 Quality Conformance Inspections - Quality conformance inspections shall consist of examinations and tests performed on the production items to determine that they meet the requirements of the following tests:

a. Individual tests (4.5.1).

b. Sampling plan tests (4.5.2).

c. Reliability Assurance Tests (4.5.3).

4.5.1 Individual Tests - Each control assembly or component, as applicable, shall be subjected to the following tests as described. under Section 4.6:

a. Displacement Gyrscope (4.6.2)

(1) Examination of product (4.6.1).

(2) Erection cycle and accuracy (4.6.2.1).

(3) Power Interruption (4.6.2.1.1).

(4) Heading rotation (4.6.2.2).

(5) Heading drift rate (4.6.2.3).

(6) Vertical drift rate (4.6.2.4). .

(7) Erection rate (4.6.2.5).

(8) Fast erection (4.6.2.5.1).

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(9) Heading precession rate (4.6.2.6).

(10) Linear acceleration sensor threshold (4.6.2.7).

(11) Turn acceleration sensor threshold (4.6.2.8).

b. Compass System Controller (4.6.3)

(1) Examination of product (4.6.1).

(2) Mode selector (4.6.3.1).

(3) Hemisphere selector (4.6.3.2).

(4) Latitude control current (4.6.3.3).

(5) Synchronization indicator deflection (4.6.3.4).

(6) Heading set operation (4.6.3.5).

(7) Timed automatic sync operation (4.6.3.5.1).

(8) Fast erection control (4.6.3.6).

(9) Visual comparison (4.6.3.7.1).

c. Electronic Control Amplifier (4.6.4).

(1) Examination of product (4.6.1).

(2) Slaving amplifier output and gain (4.6.4.1).

(3) Heading compensation range (4.6.4.2).

(4) Heading servo scale error, DG mode (4.6.4.3).

(5) Heading servo scale error, compensated, slaved mode (4.6.4.4).

(6) Roll output signal transmission (4.6.4.5).

(7) Pitch output signal transmission (4.6.4.6).

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(8) Heading servo followup rate (4.6.4.7).

(9) Heading set rate (4.6.4.10).

(10) Malfunction monitoring circuit performance (4.6.4.11).

(11) Compass or emergency mode operation (4.6.4.12).

(12) Linear acceleration control function (4.6.4.13).

(13) Roll erection and slaving control (4.6.4.14).

4.5.2 Sampling Plan Tests

4.5.2.1 Sampling Plan A - One control assembly (or component) selected at random from each 25 or less produced on the contract shall be subjected to the following tests as described under 4.6. A simulated magnetic azimuth detector may be used.

a. Individual tests (listed in 4.5.1).

b. Control assembly operational tests (4.6.5).

(1) Heading rotational polarity (4.6.5.3.1).

(2) Roll correspondence and rotational polarity (4.6.5.3.2).

(3) Pitch correspondence and rotational polarity (4.6.5.3.3).

(4) Heading set function (4.6.5.3.4).

(5) Slaving rate and correspondence (4.6.5.3.5).

(6) Control assembly heading drift rate (4.6.5.3.6).

4.5.2.2 Sampling Plan B - One control assembly (or component) selected at random from each 100 or less produced on the contract shall be subjected to the following tests as described under 4.6:

a. Sampling plan A tests (listed in 4.5.2.1).

b. Voltage and frequency variation (4.6.6).

c. Power consumption (4.6.7).

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d. Vibration error (4.6.8).

e. Low field strength (4.6.9).

f. Slaved dynamic accuracy (4.6.10).

g. Panel lighting (4.6.3.7).

4.5.2.3 Sampling Plan C - Unless otherwise specified by the procuring activity (as noted in 6.2), 3 samples of each component shall be selected at random from the first 15 items on the contract and subjected to the following applicable tests as described under 4.6:

a. Sampling plan B tests (listed in 4.5.2.2).

b. Radio noise interference and susceptibility (4.6.11).

c. Temperature-altitude (cycling)(4.6.12).

d. Humidity (4.6.13).

e. Fungus (4.6.14).

f. Rain (4.6.15).

g. Salt fog (4.6.16).

h. Dust (4.6.17).

i. Temperature shock (4.6.18).

j. Leak rate (4.6.18.1).

k. Vibration failure (4.6.19).

l. Acoustical noise (4.6.20).

m. Explosion (4.6.21).

n. Shock (4.6.22).

o. Acceleration (4.6.23).

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4.5.2.4 Rejection and Retest - When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.5.2.4.1 Individual Tests May Continue - For operational reasons, individual tests may be continued pending the investigation of a sampling test failure, but final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.2.4.2 Corrective Action and Retest - Corrective action and retest shall be required for each failure that can be reasonably attributed to poor manufacture or design practices or that adversely affects mission completion.

4.5.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by the production Acceptance Phase shall be conducted. Classification of a failure shall be in accordance with MIL-STD-781 and MIL-STD-2074.

4.5.3.1 Reliability Assurance Production Acceptance Phase - Reliability Assurance Production Acceptance Testing shall be conducted through a production burn-in. Each equipment produced but not submitted to the First Article Test shall be operated for 100 operating hours burn-in, and the test data (operating hours and failures) shall be totaled and compared to the Accept/Reject Criteria of Test Plan IV of MIL-STD-781. (Extend the lines as necessary to accommodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be available to the procuring activity. Delivery of equipment shall not be held up to await a—decision under the above test plan. In the event an Accept Decision is reached, the test duration of each equipment shall be reduced to 50 operating hours. At any time that the current total of test hours and test failure-s plotted on Test Plan IV curves (extended) show a Reject Situation, the Procuring Activity shall be notified. The Procuring Activity reserves the right to stop the acceptance of equipment at any time that a Reject Situation exists pending a review of the contractor’s efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision. For the Production Acceptance Phase, Test Level F shall be used, except that the temperature extremes shall be +55°C and -44°C, and vibration shall be 0.5G. Details of the burn-in tests shall be included in the test procedures.

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4.5.3.2 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be approved by the procuring activity prior to the beginning of the production acceptance phase of the reliability assurance tests.

4.5.3.3 Test Procedures - The procedures used for conducting reliability assurance and acceptance tests, prior to conducting tests, shall be prepared by the contractor and submitted to the procuring activity for review and approval (see 6.2.2). The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 and the applicable detailed test requirements of this section shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.5.4 Defects in Items Already Accepted - The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test Methods

4.6.1 Examination of Product - The control assemblies and components shall be examined to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.2 Displacement Gyroscope - The displacement gyroscope shall be tested as follows:

a. Connect stator leads of heading, pitch, and roll servoed repeaters to the outputs of the connector pins of the displacement gyroscope as shown in table XII:

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TABLE XII. Servoed Repeater Stator Connections to Displacement Gyroscope

Heading	Pitch	Roll
S1 to 1 S2 to 2 S3 to 3	S1 to 10 S2 to 11 S3 to 12	S1 to 13 S2 to 14 S3 to 15

b. Connect a jumper between the displacement gyroscope connector 1J1-7 and 1J1-5.

c. Excite rotors of all servoed repeaters from 400 Hz, øA with R1 high and R2 low.

d. Connect 400 Hz input power through a normally Open disconnect means to connector (1J1) pins as follows:

(1) Neutral to 1J1-21.

(2) 115V øA to 1J1-22.

(3) 115V øB to 1J1-23.

(4) 115V øC to 1J1-24.

(5) 6.5V øC to 1J1-6.

4.6.2.1 Erection Cycle and Accuracy - The displacement gyroscope shall be so aligned that the mounting surface will be parallel to a horizontal surface within 0.05° and power shall then be applied. After 1 minute, the displacement gyroscope shall be subjected to scorsby motion. Erection shall be completed in 2-1/2 minutes in pitch and in roll after which the pitch and roll attitude excursions shall be noted and the average pitch and roll indications shall be computed. The average pitch and roll indications shall be 00 within ±1/40. The scorsby motion shall be turned off.

4.6.2.1.1 Power Interruption - The displacement gyroscope shall be level within 20 and shall have its power removed for 25 seconds. During this period, the displacement gyroscope shall be displaced through angles of ±60° to ±65° in pitch and in roll and then returned to its original

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position. Within 10 seconds after reapplication of power, the pitch and roll axes shall indicate the original position to with ±1°. After 2 1/2 minutes, the pitch and roll axes shall indicate the original position within ±1/4°.

4.6.2.2 Heading Rotation - The case shall be rotated about the vertical axis in the direction of a right turn. The heading indicator shall show an increasing angle.

4.6.2.3 Heading Draft Rate - The displacement gyroscope shall be subjected to scorsby motion and the initial heading and the heading at the end of each 3 consecutive 1/2 hour intervals shall be recorded. The rate of heading change shall be computed in degrees per hour for each interval and the results adjusted to obtain free drift rate by adding the earth's rate correction for the latitude of test. The earth's rate correction is +15 sine latitude degrees per hour (Northern Hemisphere). The results shall be as follows:

a. The average of the three hourly drift rates shall not exceed ±1/20 per hour. (For the tests specified in 4.4.1.1 and 4.6.12 through 4.6.24, the normal limit shall be ±1° per hour.)

b. The spread between the most positive and the most negative hourly rates shall not exceed 1° per hour.

4.6.2.4 Vertical Drift Rate - The displacement gyroscope shall be subjected to scorsby motion and tested as follows:

a. Disconnect pitch erection by opening the jumper between connector 1J1-7 and 1J1-5 and simultaneously record pitch attitude. Record pitch attitude at the end of a 15 minute interval and compute rate of pitch change in degrees per hour. Adjust the result to obtain free drift rate by adding the earth's rate correction. The earth's rate correction in pitch is +15 cosine latitude sine H degrees per hour, where H is the heading of the displacement gyroscope fore-art axis with respect to geographic North. (H = 0° for North, 90° for East). The pitch drift rate shall not exceed 12° per hour. Reconnect the jumper between connector 1J1-7 and 1J1-5.

b. Disconnect roll erection by removing the 6.5V øC from connector 1J1-6 and simultaneously record roll attitude. Record roll attitude at the end of a 15 minute interval and compute rate of roll change in degrees per hour. Adjust the result to obtain free drift rate by adding the earth's rate correction. The earth's rate correction in roll is -15 cosine latitude cosine H degrees per hour. The roll drift rate shall not exceed 12° per hour. Reconnect the 6.5V øC to connector 1J1-6.

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4.6.2.5 Erection Rate - During the erection rate measurement interval, the displacement gyroscope shall be subjected to scorsby motion and tested as follows:

a. Disconnect the roll axis servoed repeater and apply an isolated; center-tapped, 11.8V, øA voltage to connector 1J1-13, 1J1-14, and 1J1-15 with the high side on 15 and the center tap on 14.

b. Disconnect pitch erection by opening the jumper between connector 1J1-7 and 1J1-5.

c. Reverse pitch erection polarity by connecting 26V negative øC to connector 1J1-5. Remove this negative øC voltage when the pitch attitude becomes ±10° ±1° and remove the isolated 11.8V øA voltage from connector 1J1-13,–1J1-14, and 1J1-15.

d. Reapply pitch erection by reconnecting the jumper between connector 1J1-7 and 1J1-5, and begin timing when the pitch attitude becomes ±8°. Note the pitch attitude at the end of 1 minute and compute the pitch erection rate by correcting for earth's rotation per 4.6.2.4a. The pitch erection rate shall be 40 to 60 per minute.

e. Reconnect the roll axis servoed repeater.

f. Disconnect the pitch axis servoed repeater, and apply an isolated center taped 11.8V øA voltage to connector 1J1-10, 1J1-11, and 1J1-12 with the high side on 12 and the center tap on 11.

g. Disconnect roll erection by removing the 6.5V øC from connector 1J1-6.

h. Reverse roll erection polarity by connecting 26V negative øC to connector 1J1-6. Remove this negative øC voltage when the roll attitude becomes ±1O ±1° and remove the isolated 11.8V øA voltage from connector 1J1-1O, 1J1-11, and 1J1-12.

i. Reapply roll erection by reconnecting 6.5V øC to connector 1J1-6 and begin timing when the roll attitude becomes ±8°. Note the roll attitude at the end of 1 minute, and compute the roll erection rate by correcting for the earth's rotation per 4.6.2.4b. the roll erection rate shall be 0.75° to 1.75° per minute.

j. Reconnect the pitch axis servoed repeater.

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4.6.2.5.1 Fast Erection - The procedure of 4.6.2.5 shall be repeated except:

a. In steps c and h, tilt to attitude angles of ±32 ±l°.

b. In step d, apply pitch erection by connecting a jumper between connector 1J1-8 and 1J1-5. Replace by a jumper between connector 1J1-7 and 1J1-5 upon completion of this step.

c. In step i, apply roll erection by connecting a jumper between connector 1J1-8 and 1J1-6. Replace the 6.5V øC to connector 1J1-6 upon completion of this step.

d. In steps and i, begin timing at attitude angles of ±30°. The erection rate shall at least 12° per minute.

4.6.2.6 Heading Precession Rate - The displacement gyroscope shall be subjected to scorsby motion and tested as follows:

a. Apply +1.100 ±0.005 mA dc with positive on North latitude control connector 1J1-17 and negative on ground 1J1-21. Determine the rate of heading change during a 30 minute interval and correct the result for the earth's rotation to obtain the heading precession rate.

(1) The heading precession rate shall be +9 ±3/4° per hour. (For the tests specified in 4.4.1.1 and 4.6.12 through 4.6.24, the normal limit shall be +9 ±1 1/4° per hour.)

b. Disconnect the dc input from connector 1J1-17.

c. Apply +11.0 ±0.1 ma dc with positive on South latitude control connector 1J1-18 and negative on ground connector 1J1-21. Determine the rate of heading change during a 6 minute interval and correct the result for the earth's rotation to obtain the heading precession rate.

(1) The heading precession rate shall be -90 ±20± per hour.

d. Disconnect the dc input from connector 1J1-18.

4.6.2.7 Linear Acceleration Sensor Threshold - Connect 26 ±0.25v, 400 Hz to connector 1J1-16 through a 6,000 ±1% ohm resistor. Measure the voltage from connector 1J1-16 to 1J1-21 using a high impedance ACVM and test as follows:

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a. Rotate the gimbal system in positive pitch by following the sequence defined in 4.6.2.5a, b, and c, except remove the test voltages per 4.6.2.5c when the pitch angle has become approximately So.

b. Reapply pitch erection and monitor the pitch angle and the voltage indicated by the ACVM. (Pitch erection may be applied intermittently to facilitate testing.)

(1) The voltage shall increase to 12Vac at a positive pitch angle of 4° 15′ or less. Record the angle.

(2) The voltage shall increase to 15Vac at a positive pitch angle of 3° 15′ or greater. Record the angle.

(3) The voltage shall increase to 21Vac minimum when the displacement gyroscope is fully erected in pitch. Record the voltage.

c. Fully erect the displacement gyroscope in pitch.

d. Rotate the gimbal system in negative pitch by following the sequence defined in 4.6.2.5a, b, and c, except reverse the polarity of the isolated 11.8V øA connection in 4.6.2.5a and remove the test voltages per 4.6.2.5c when the pitch angle has become approximately So.

e. Reapply pitch erection and monitor the pitch angle and the voltage indicated by the ACVM. (Pitch erection may be aplied intermittently to facilitate testing.)

(1) The voltage shall increase to 12Vac at a negative pitch angle of 4° 15′ or less. Record the angle.

(2) The voltage shall increase to 15Vac at a negative pitch angle of 3° 15′ or greater. Record the angle.

f. Fully erect the displacement gyroscope in pitch.

4.6.2.7.1 Reliability and Environmental Test Limits - When the test specified in 4.6.2.7 is performed during the tests specified in 4.4.1.1 and 4.6.12 through 4.6.24, the following shall apply:

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a. In 4.6.2.7b(1) and e(1), the positive pitch angle and the negative pitch angle, respectively, shall be 4° 30′ in lieu of 4° 15′ .

b. In 4.6.2.7b(2) and e(2), the positive pitch angle and the negative pitch angle, respectively, shall be 3° 00′ in lieu of 30 15′ .

4.6.2.8 Turn Acceleration Sensor Threshold - Connect 26 ±0.25V, 400 Hz to connector 1J1-9 through a 6,000 ±1% ohm resistor. Measure the ac voltage from connector 1J1-9 to 1J1-21 using a high impedance ACVM and test as follows:

a. Rotate the gimbal system in positive roll by following the sequence defined in 4.6.2.5f, g, and h, except remove the test voltages per 4.6.2.5h when the roll angle has become approximately 5°.

b. Reapply roll erection and monitor the roll angle and the voltage indicated by the ACVM. (Roll erection may be applied intermittently to facilitate testing.)

(1) The voltage shall increase to 12Vac at a positive roll angle of 4° 15′ or lesss Record the angle.

(2) The voltage shall increase to 15Vac at a positive roll angle of 3° 15′ or greater. Record the angle.

(3) The voltage shall increase to 21Vac minimum when the displacement gyroscope is fully erected in roll. Record the voltage.

c. Fully erect the displacement gyroscope in roll.

d. Rotate the gimbal system in negative roll by following the sequence defined in 4.6.2.5f, g, and h, except reverse the polarity of the isolated 11.8V øA connection in 4.6.2.5f and remove the test voltages per 4.6.2.5h when the roll angle has become approximately 50.

e. Reapply roll erection and monitor the roll angle and the voltage indicated by the ACVM. (Roll erection may be applied intermittently to facilitate testing.)

(1) The voltage shall increase to 12Vac at a negative roll angle of 40 15′ or less. Record the angle.

(2) The voltage shall increase to 15Vac at a negative roll angle of 3° 15′ or greater. Record the angle.

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f. Fully erect the displacement gyroscope in roll.

4.6.2.8.1 Reliability and Environmental Test Limits - When the test specified in 4.6.2.8 is performed during the tests specified in 4.4.1.1 and 4.6.12 through 4.6.24, the following shall apply:

a. In 4.6.2.8b(1) and e(1), the positive and negative roll angle, respectively, shall be 4° 30′ in lieu of 4° 15′.

b. In 4.6.2.8b(2) and e(2) the positive and negative roll angle, respectively, shall be 3° 00′ in lieu of 3° 15′.

c. Disconnect all test equipment.

4.6.3 Compass System Controller - The following compass system controller tests shall be conducted.

4.6.3.1 Mode Selector - The mode selector switch on the compass system controller shall be tested as follows:

a. Measure the resistance between connector 3J1-5 (Hi) or 3J1-24 (Hi), and 3J1-23 (LOW). The resistance shall be within the following limits shown in table XIII for each position of the mode selector switch:

TABLE XIII. Mode Selector Switch Resistance Tests

Mode Selector	(3J1) Pins	Resistance
DG	5 and 23	1 megohm min
DG	24 and 23	1 megohm min
SLAVED	5 and 23	1 ohm max
SLAVED	24 and 23	1 megohm min
COMP or EMERG	5 and 23	1 megohm min
COMP or EMERG	24 and 23	1 ohm max

4.6.3.2 Hemisphere Selector - The hemisphere selector switch on the compass system controller shall be tested as follows:

a. Set the latitude control at the 900 position.

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b. Measure the resistance between connector 3J1-9 and 3J1-10. The resistance shall be within the following limits for each position of the hemisphere switch:

Hemisphere Switch	Resistance
N	20 kilohms max
S	1 megohm min

c. Measure the resistance between connector 3J1-8 and 3J1-10. The resistance shall be within the following limits for each position of the hemisphere switch:

Hemisphere Switch	Resistance
N	1 megohm min
S	20 kilohms max

4.6.3.3 Latitude Control Current - One 1,000 ±0.1% ohm resistor shall be connected from connector 3J1-8 to 3J1-10 and another from connector 3J1-9 to 3J1-10. Apply +27 ±1Vdc with negative on connector 3J1-10 and positive in series with an 820 ±0.2% ohm resistor to connector 3J1-4. The following shall then be accomplished:

a. Set the hemisphere selector switch in N position and connect an adjustable resistance unit having a 0.1% accuracy from +27Vdc to connector 3J1-8. Set the latitude control on the compass system controller and the adjustable resistance to the values shown in table XIV and read dc voltage from connector 3J1-8 to 3J1-9. The voltage must not exceed ±30 millivolts.

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TABLE XIV. Latitude Control (N) Resistance Settings

Latitude Control (Degrees)	Resistance (Ohms)
00.0	open
05.0	167,750
10.0	83,906
15.0	55,962
20.0	42,062
25.0	33,839
30.0	28,444
35.0	24,665
40.0	21,920
45.0	19,833
50.0	18,231
60.0	16,003
70.0	14,670
80.0	13,958
90.0	13,730

b. Set the hemisphere selector switch to the S position and change the connector 3J1-8 end of the adjustable resistance unit to connector 3J1-9. Set the latitude control and ajustable resistance to the values shown in table XV and read dc voltage from connector 3J1-S to 3J1-9. the voltage must not exceed ±30 millivolts.

TABLE XV. Latitude Control (S) Resistance Settings

Latitude Control (Degrees)	Resistance (Ohms)
00.0	open
30.0	28,444
90.0	13,730

c. Disconnect all test equipment.

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4.6.3.4 Synchronization Indicator Deflection - The following test shall be conducted for the sync indicator on the compass system controller.

a. Apply +27 ±1Vdc through a 56.2 kohm resistor with positive on connector 3J1-6 and negative on 3J1-7.

(1) The sync indicator pointer must deflect halfway toward plus (+).

b. Repeat step a, with -27 ±1Vdc polarity with deflection halfway toward minus (-).

c. Disconnect all test equipment.

(1) The sync indicator pointer must return to center within ±1/16 inch.

4.6.3.5 Heading Set Operation - A voltage and frequency of 26 ±1V, 400 Hz shall be connected from connector 3J1-3 to 3J1-23 (grid). 'For the compass system controller with the timed automatic synchronization specified in 3.8.3.2 the mode selector switch must be in either the COMP or EMERG position. The following test shall be conducted:

a. The resistance from connector 3J1-1 (Hi) to 3J1-23 (Low) shall be measured. Resistance shall be within the following limits for each position of the heading set control:

Heading Set	Resistance
Depressed	1 ohm maximum
Released	1 megohm minimum

b. For the compass system controller with fast erection combined with the heading set control as specified in 3.8.3a repeat step a, measuring resistance from connector 3J1-12 to 3J1-23.

c. Measure ac voltage from connector 3J1-2 to 3J1-23. Voltage shall be within the following limits for each position of the heading set control:

Heading Set	Voltage
Depressed & full cw	4 to 5Vac
Depressed & full ccw	4 to 5Vac
Released	75 mV max

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d. Measure ac voltage from connector 3J1-3 to 3J1-2 with the heading set control depressed and rotated fully ccw.

(1) Voltage shall be 30.5 ±2Vac.

e. Reduce applied voltage to 2.5 ±0.2Vac.

(1) Output voltage from connector 3J1-2 to 3J1-23 (grid) shall be less than 150 mV when heading set control knob is depressed and rotated full cw and full ccw.

4.6.3.5.1 Timed Automatic Sync Operation - For the compass system controller with the timed automatic synchronization circuit specified in 3.8.3.2, make connections and perform the tests as follows:

a. Apply 26 ±0.5V, 400 Hz from connector 3J1-3 to 3J1-23.

b. Apply 28 ±2Vdc through a resistance of 1,000 ±5% ohms with the high side on connector 3J1-1 and the low side on 3Jl-23.

c. Set the mode selector switch to EMERG and measure the dc voltage from connector 3J1-1 to 3J1-23.

(1) The voltage shall be greater than 25Vdc.

d. Rotate mode selector switch from EMERG to SLAVED and begin timing.

(1) The voltage from connector 3J1-1 to 3J1-23 shall be less than 2Vac for 10 ±2 seconds, then shall rise to a value greater than 25Vac.

e. Rotate mode selector switch to DG.

(1) The voltage shall be greater than 25Vac.

f. Rotate mode selector switch from DG to SLAVED and begin timing.

(1) The voltage shall be less than 2Vac for 10 ±2 seconds, then shall rise to greater than 25Vac.

g. Disconnect all test equipment.

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4.6.3.6 Fast Erection Control - For the compass system controller with the separate fast erection function specified in 3.8.3.3.b connect an ohmmeter between connector 3J1-12 to 3J1-23. The resistance shall be measured for each position of the ERECT switch as follows:

ERECT Switch	Resistance
Depressed	1 ohm maximum
Released	1 megohm minimum

a. Disconnect all test equipment.

4.6.3.7 Lighting - On production equipment, the following compass system controller test may be conducted prior to final assembly:

a. Apply the following ac or dc voltages between connector 3J1-21 and 3J1-22:

(1) DCK-128/A24G	4.50 ±.05V
(2) DCK-129/A24G	5.0 ±0.1V
(3) DCK-228/A24G	5.0 ±0.1V
(4) DCK-234/A24G	4.50 ±0.05V
(5) DCK-235/A24G	5.0 ±0.1V
(6) DCK-236/A24G	5.0 ±0.1V

b. Brightness and color shall be checked to determine compliance with 3.8.

c. When this test is performed under First Article or Sampling Plan B, the contrast, stray light, and dial visibility shall meet the requirements of 3.8.

d. Remove all test connections.

4.6.3.7.1 Visual Comparison - The prime standard and the compass system controller under test shall be placed side-by-side at eye level. The two instrument panels shall be viewed at a distance of 2 ±1/2 feet in a sighting cone 30° from normal to the surface of the panel. With the applicable voltage specified in 3.8 applied to the lighting pins the overall lighting shall be compared. With 2.85Vdc applied to the prime overall standard and the compass system controller under test, the general overall lighting shall again be compared. The questionable areas of production units failing this test shall be measured for color and brightness as specified in the applicable lighting specification (see

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3.8). If the overall lighting does not fall within the color and brightness limits of the applicable lighting specification, the compass system controller shall be rejected. This test shall be conducted in a completely dark room.

4.6.4 Electronic Control Amplifier Tests - The following connections are necessary to perform the electronic control amplifier test:

a. Connect the simulated controller to the electronic control amplifier using the compass system controller connections (2J3) shown in table IV.

b. Connect the simulated displacement gyroscope heading transmitter stator leads to the following connector (2J2) pins as shown in table III:

S1 to 1
S2 to 2
S3 to 3

c. Connect the simulated magnetic azimuth detector stator leads to the following connector (2J3) pins as shown in table IV:

S1 to 16
S2 to 17
S3 to 18

d. Connect the servoed repeater to each heading output as required using the following connector (2J1) pins as shown in table V. Output No. 1 is to be monitored unless specifically stated otherwise.

Output No. 1	Output No. 2	Output No. 3	Output No. 4
S1 to 1	S1 to 4	S1 to 7	S1 to 10
S2 to 2	S2 to 5	S2 to 8	S2 to 11
S3 to 3	S3 to 6	S3 to 9	S3 to 12
R1 to 13	R1 to 15	R1 to 17	R1 to 19
R2 to 14	R2 to 16	R2 to 18	K2 to 20

e. Connect the simulated displacement gyroscope heading transmitter stator leads to the following connector (2J2) pins as shown in table III during pitch and roll channel tests.

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For Pitch Channel Tests

S1 to 19
S2 to 20
S3 to 21

For Roll Channel Tests

S1 to 22
S2 to 23
S3 to 24

f. Connect a 1000 ±0.1% ohm resistor between each of the following connector (2J2) pins–per table III:

From 17 to 9
From 18 to 9

g. Connect 400 Hz input power to the following connector (2J1) pins per table V:

Neutral to 23
115V øA to 24
115V øB to 25
115V øC to 26

h. Connect 78V øC through a spring-loaded normally open switch to connector 2J2-4 with the neutral (unstitched) to 2J2-9 or equivalent ground.

4.6.4.1 SlavingAmplifier Output and Gain - The slaving amplifier output and gain shall be checked as follows:

a. Set the controls on the simulated controller and electronic control amplifier as follows:

Mode switch to SLAVED
Latitude control to 0°
Hemisphere switch to N
Compensation switch to OFF

b. Set the simulated azimuth detector to 0° and the simulated displacement gyroscope heading transmitter to 0°.

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c. Synchronize the electronic control amplifier by depressing and holding the heading set control knob on the controller until the servoed repeater stops rotating.

d. Note the deflection of the sync indicator pointer on the controller, and measure the dc voltage with poisitive on connector 2J2-18 and negative on 2J2 as noted in table III.

(1) The sync indicator pointer must be centered within ±1/16 inch.

(2) The dc voltage from connector 2J2-18 to 2J2-17 must be 0 ±1Vdc.

e. Test for slaving amplifier (positive) output as follows:

(1) For the ASK-32/A24G-39 electronic control amplifier, position the displacement gyroscope heading transmitter to the following headings shown in table XVI. The sync indicator pointer on the controller must deflect toward positive (+) to the proportion of scale noted and the dc voltage must agree with the tabulation shown in table XVI.

TABLE XVI. Sync Indicator Deflection Tests

Heading Transmitter	Deflection	volts
0° 15′	Part scale	---
30	---	+11 ±2Vdc
40	Full scale	---
90°	Full scale	+11 ±3Vdc

(2) For the ASK-44/A24G electronic control amplifier position the displacement gyroscope heading transmitter to the headings shown in table XVII. The sync indicator pointer on the controller must deflect toward positive (+) to the proportion of scale noted, and the dc voltage must agree with the tabulation shown in table XVII.

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TABLE XVII. Sync Indicator Deflection Tests

Heading Transmitter	Deflection	Volts
0° 30' 6° 8° 90°	Part scale --- Full scale Full scale	--- +11 ±2Vdc --- +11 ±3Vdc

f. Position the azimuth detector to 90° and re-synchronize the system, leaving the displacement gyroscope heading transmitter at 90°.

(1) The sync indicator pointer must be centered within ±1/16 inch.

(2) The dc voltage from connector 2J2-18 to 2J2-17 must be 0 ±1Vdc.

g. Test for slaving amplifier (negative) output as follows:

(1) For the ASK-32/A24G-39 electronic control amplifier, position the simulated azimuth detector to the following headings. The sync indicator pointer on the controller must deflect toward negative (-) to the proportion of scale noted, and the dc voltage must agree with the following tabulation shown in table XVIII.

TABLE XVIII. Sync Indicator Deflection Tests

Azimuth Detector Heading	Deflection	volts
90° 15' 93° 94° 180°	Part scale --- Full scale Full scale	--- -11 ±2Vdc - --- -11 ±3Vdc

(2) For the ASK-44/A24G electronic control amplifier, position the simulated azimuth detector to the following headings. The sync indicator pointer on the controller must reflect toward negative (-) to the proportion of scale noted, and the dc voltage must agree with the following tabulation shown in table XIX.

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TABLE XIX. Sync Indicator Deflection Tests

Azimuth Detector Heading	Deflection	Volts
90° 30'	Part scale	---
96°	---	-11 ±2Vdc
98°	Full scale	---
180°	Full scale	-11 ±3Vdc

h. Turn the mode selector on the controller to DG mode.

(1) The sync indicator pointer on the controller must become centered within ±1/16 inch.

(2) The dc voltage from connector 2J2-18 to 2J2-17 must become 0 ±1Vdc.

i. Set the latitude control on the controller to 90.00.

(1) The sync indicator pointer on the controller must remain centered within ±1/16 inch.

(2) The dc voltage from connector 2J2-18 to 2J2-17 must become -1.833 ±0.050Vdc.

4.6.4.2 Heading Compensation Range - The heading compensation range shall be checked as follows:

a. Set the controls on the controller and electronic control amplifier as follows:

Mode switch to SLAVED
Latitude control to 0°
Hemisphere switch to N
Compensation switch to OFF

b. Set all compensation potentiometers on the electronic control amplifier approximately at center position.

c. Turn compensation switch on the electronic control amplifier to the ON position.

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d. Rotate 0° compensation potentiometer fully cw. Synchronize the electronic control amplifier by depressing and holding the heading set control on the controller, then position the azimuth detector to obtain a 0° heading.

e. Set the compensation switch to the OFF position and re-synchronize the electronic control amplifier.

(1) The heading must have decreased by 4 ±1° from the compensated value.

f. Set the compensation switch to the ON position.

g. Rotate the 0° compensation potentiometer on the electronic control amplifier fully ccw. Re-synchronize the electronic control amplifier and position the azimuth detector to obtain a 0° heading.

h. Set the compensation switch to the OFF position and re-synchronize the electronic control amplifier.

(1) The heading must have increased by 4 ±1° from the compensated value.

f. Return the 0° compensation potentiometer on the electronic control amplifier approximately to the center position.

j. Repeat steps c, d, e, f, g, h, and i with the heading set at each 15° and with the correspondingly marked compensation potentiometer being employed.

4.6.4.3 Heading Servo Scale Error, IX Mode - The heading servo scale error DG mode shall be checked as follows:

a. Set the controls on the controller and electronic control amplifier as follows:

Mode switch to IX
Latitude control to 0°
Hemisphere switch to N
Compensation switch to OFF

b. Set the displacement gyroscope heading transmitter to 0°.

c. Depress and rotate heading set control knob on the controller to position the servoed repeater connected to heading output No. 1 to 0°.

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d. Check the heading indicated by output No's 2, 3, and 4. Each indicated heading shall be 0° °6'.

e. Position the displacement gyroscope heading transmitter to each 30° beginning at 0° and record the indicated heading for output No. 1.

f. Determine the point errors by subtracting the displacement gyroscope heading from the indicated heading at each position.

g. Calculate the scale error spread by determining the maximum algebraic difference between the point errors. The scale error spread shall not exceed 30 minutes.

h. Repeat steps e, f, and g for heading output No's 2, 3, 4.

(1) The scale error spread for output No's 2 and 3 shall not exceed 30 minutes.

(2) The scale error spread for output No. 4 shall not exceed 36 minutes.

4.6.4.3.1 Reliability and Environmental Test Limits - When the test specified in 4.6.4.3 is performed during the tests specified in 4.4.1.1 and 4.6.12 through 4.6.24, the indicated heading specified in 4.6.4.3d shall be 0° ±9' in lieu of 0° ±6'.

4.6.4.4 Heading Servo Scale Error, Compensated, Slaved Mode - The compensated heading servo scale error in the slaved mode shall be checked as follows:

a. Set the controls on the controller and the electronic control amplifier as follows:

Mode switch to SLAVED
Latitude control to O°
Hemisphere stitch to N
Compensation switch to ON

b. Set all compensation potentiometers on the electronic control amplifier approximate at center position.

c. Set the simulated magnetic azimuth detector heading to 0° 23' in accordance with the first row of the tabulation shown in table XVIII under step f.

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d. Set the indicated heading of output No. 1 to 0° as tabulated, using the simulated displacement gyroscope heading transmitter.

e. Adjust the 0° compensation potentiometer on the electronic control amplifier to produce a null voltage from connector 2J2-17 to 2J2-18 as follows:

ASK-32/A24G-39	0 ±0.5 Vdc
ASK-44/A24G	0 ±0.25 Vdc

f. Repeat steps c, d, and e for each of the table XX tabulated heading pairs, in sequence.

TABLE XX. Heading Scale Error Tabulation

Azimuth Detector Heading	Indicated Heading
0° 23'	0°
16° 09'	15°
31° 49′	30°
47° 23'	45°
62° 46'	60°
77° 58′	75°
92° 58'	90°
107° 46′	105°
122° 23′	120°
136° 49′	135°
151° 09′	15o°
165° 23′	165°
179° 37′	180°
193° 51′	195°
208° 11′	210°
222° 37′	225°
237° 14′	240°
252° 02′	255°
267° 02′	270°
282° 14′	285°
297° 37′	300°
313° 11′	315°
328° 51′	330°
344° 37′	345°

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g. A second compensation is permitted beginning with the compensation potentiometers on the electronic control amplifier positioned as in e and f (this would consist of repeating steps c, d, e, and f).

h. Reset the azimuth detector to each heading as tabulated, and set the indicated heading to produce a null voltage as follows from connector 2J2-17 to 2J2-18. The indicated heading shall agree with the tabulation within +9 minutes.

ASK-32/A24G-39	O ±0.5 Vdc
ASK-441A24G	0 ±0.25 Vdc

i. Recenter the compensation potentiometers on the electronic control amplifier and turn the compensation switch on the electronic control amplifier to the OFF position.

4.6.4.5 Roll Output Signal Transmission - The roll output signal transmission test shall be conducted as follows:

a. Set the displacement gyroscope roll transmitter to O°; the indicated roll output at connector 2J4 shall be 0° ±6′.

b. Position the displacement gyroscope roll transmitter to 90°; the indicated roll output shall be 90° ±6'.

4.6.4.6 Pitch Output signal Transmission - the test specified in 4.6.4.5 shall be conducted except the displacement gyroscope pitch transmitter shall be used and the indicated pitch output shall be monitored.

4.6.4.7 Heading Servo followup Rate - the heading servo followup rate shall be checked as follows:

a. Set the controls on the controller and the electronic control amplifier as follows:

Mode switch to DG
Latitude control to 0°
Hemisphere switch to N
Compensation switch to OFF

b. Rotate the displacement gyroscope heading transmitter at a rate of 60° per second minimum. The indicated heading shall rotate smoothly.

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4.6.4.8 Roll Servo Followup Rate - Deleted.

4.6.4.9 Pitch Servo Followup Rate - Deleted.

4.6.4.10 Heading Set Rate - The heading set rate test shall be conducted as follows:

a. Set the controls on the controller and the electronic control amplifier as follows:

Mode switch to SLAVED
Latitude control to 0°
Hemisphere switch to N
Compensation switch to OFF

b. Set the azimuth detector to 0°.

c. Depress and hold the heading set control knob on the controller until the indicated heading becomes synchronized with the azimuth detector heading. Record the indicated heading.

d. Hold the heading set control knob depressed but not deflected and rotate the displacement gyroscope heading transmitter in an increasing heading direction at a rate of 30° per second. The indicated heading shall increase from the value recorded in step c by the following amount:

ASK-321A24G-39	by no more than 5°
AsK-44/A24G	by no more than 30°

e. Discontinue rotation of the displacement gyroscope heading transmitter and release the heading set control knob.

f. Turn the mode stitch on the controller to DC mode.

g. Depress the heading set control knob on the controller and rotate it cw to contact the stop. The indicated heading shall increase at 30° per second minimum.

h. Measure resistance from connector 2J3-13 to 2J3-15 and 2J3-14 to 2J3-15 with the heading set control knob depressed and with it released. Resistance shall be within the limits shown in table XXI.

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TABLE XXI. Heading Set Control Resistance Measurements

Heading Set Control Position	2J3-13 to 2J3-15	2J3-14 to 2J3-15
Depressed Released	1 ohm max 1 megohm min	1 megohm min 1 ohm max

4.6.4.11 Malfunction Monitoring Circuit - The malfunction monitoring circuit shall be tested as follows:

a. Set the controls on the controller and the electronic control amplifier as follows:

Mode switch to DG
Latitude control to O°
Hemisphere switch to N
Compensation switch to OFF

b. Increase the input supply voltage (keeping all phases balanced) until a malfunction is signaled; then repeat by increasing the input supply voltage independently on each phase until a malfunction is signaled (Note: do not exceed 138V line-to-nautral).

(1) A malfunction shall be signaled as specified in 3.7.6.1, for a balanced input supply voltage in the range of 126.5 to 138V line-to-neutral.

(2) A malfunction shall be signaled in accordance with 3.7.6.1, for a øA, øB, or øC input supply voltage in the range of 126.5 to 138V line-to-neutral.

(3) Return the input supply voltage to 115 ±1V line-to-neutral. The malfunction signals shall disappear.

c, Decrease the input supply voltage (keeping all phases balanced) until a malfunction is signaled; then repeat by decreasing the input supply voltage independently On each phase until a malfunction is signaled.

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(1) A malfunction shall be signaled as specified in 3.7.6.1, for a balanced input supply voltage in the range of 103.5 to 92V line-to-neutral.

(2) A malfunction shall be signaled in accordance with 3.7.6.1, for a øA, øB, or øC input supply voltage in the range of 103.5 to 92V line-to-neutral.

d. Return the input supply voltage to 115 ±1V line-to-neutral. The malfunction signals shall disappear.

e. Position the displacement gyroscope heading transmitter to 0°.

f. Set the indicated heading to 0°.

g. Prevent the heading servo motor from rotating.

h. Slowly rotate the displacement gyroscope heading transmitter in a positive direction until a malfunction is signaled as specified in 3.7.6.1. A malfunction shall be signaled within the region of +2° +3°.

i. Slowly return the displacement gyroscope heading transmitter to 0°. The malfunction signals shall have disappeared when the displacement gyroscope heading becomes +2°.

j. Return heading servo loop to condition prior to step g.

k. Remove the excitation from the displacement gyroscope heading transmitter. A malfunction shall be signaled as specified in 3.7.6.1.

l. Reconnect the displacement gyroscope heading transmitter excitation. The malfunction signals shall disappear.

m. Disconnect the displacement gyroscope heading transmitter stator leads in pairs from connector 2J2-1, 2J2-2, and 2J2-3.

(1) A malfunction shall be signaled as specified in 3.7.6.1.

(2) The malfunction signals shall disappear with all leads connected.

n. Apply 78V ±C to connector 2J2-4 by closing spring-loaded switch per 4.6.4.g.

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(1) A malfunction shall be signaled as specified in 3.7.6.2.

(2) Turn mode selector switch on controller to the SLAVED position. The malfunction signal shall remain.

(3) Turn mode selector switch on controller to the COMP or EMERG position. The malfunction signals shall disappear.

4.6.4.12 Compass or Emergency Mode Operation - Compass or emergency mode operation shall be checked as follows:

a. Set the controls on the controller and the electronic control amplifier as follows:

Mode switch to COMP or EMERG
Latitude control to 0°
Hemisphere switch to N
Compensation switch to OFF

b. Rotate the azimuth detector in 30° increments noting the servoed repeater heading output. The difference between the azimuth detector position and the servoed repeater heading output shall not vary by more than 1° at each point.

4.6.4.13 Linear Acceleration Control Functions - The linear acceleration control functions shall be checked as follows:

a. Set the controls on the controller and the electronic control amplifier as follows:

Mode switch to SLAVED
Latitude Control to 0°
Hemisphere switch to N
Compensation switch to OFF

b. Set a decade resistor to 8,000 ohms and connect it from connector 2J2-5 to 2J2-9.

C. Position the azimuth detector to O°.

d. Depress and hold the heading set control knob on the controller until the indicated servoed repeater heading becomes synchronized with the azimuth detector heading.

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e. Reposition the azimuth detector to 10°.

f. Note the deflection of the sync indicator pointer on the controller. The sync indicator pointer on the controller shall be at negative (-) full scale.

g. Measure ac voltages at the tabulated pins of connector (2J2). Voltages shall be within the following limits:

2J2 Pins	volts
7 to 9	26 ±1Vac
7 to 12	89 ±2Vac

h. Set the decade resistor to 5,000 ohms.

(1) The voltage from connector 2J2-7 to 2J2-9 must become 6.5 ±1Vac.

(2) For the ASK-32/A24G-39 the sync indicator pointer on the controller shall become centered within ±1/16 inch.

(3) For the ASK-44/A24G the sync indicator pointer on the controller shall remain at negative (-) full scale.

i. Set the decade resistor to 8,000 ohms.

(1) The ac voltage from connector 2J2-7 to 2J2-9 must return to 26 ±1Vac.

(2) For the ASK-32/A24G-39 the sync indicator pointer on the controller shall return to negative (-) full scale.

(3) For the AsK-44/A24G the sync indicator pointer on the controller shall remain at negative (-) full scale.

4.6.4.14 Roll Erection and Slaving Control

4.6.4.14.1 ASK–32/A24G-39 Electronic Control Amplifier - The roll erection and slaving control for the ASK–32/A24G-39 shall be checked as follows:

a. Set the controls on the controller and the electronic control amplifier as follows:

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Mode switch to SLAVED
Latitude control to 0°
Hemisphere switch to N
Compensation switch to OFF

b. Position the azimuth detector to 0°.

c. Depress and hold the heading set control knob on the controller until the indicated servoed repeater heading becomes synchronized with the azimuth detector.

d. Reposition the azimuth detector to 340°.

e. Measure the dc voltage from connector 2J2-17 to 2J2-18. The dc voltage shall be greater than 8.0Vdc.

f. Measure ac voltages at the tabulated pins of connector 2J2. Voltages shall be within the following limits:

(2J2) Pins	Volts
8 to 9	6.5 ±1Vac
8 to 12	108.5 ±2Vac

g. Rotate the displacement gyroscope heading transmitter at a rate of 3° per minute for 1.5 minutes.

(1) The voltage from connector 2J2-8 to 2J2-9 shall remain at 6.5 ±1Vac.

(2) The voltage from connector 2J2-17 to 2J2-18 shall remain greater than 8Vdc.

h. Repeat step g with a rate of -3° per minute for 1.5 minutes.

4.6.4.14.1.1 Switching Rate - The switching rate shall be checked as follows. (Note: During steps b through d, the dc voltage may temporarily become less than 1Vdc while the switching rate is applied. The time at which this occurs shall not be recorded.)

a. Set the controls on the controller and the electronic control amplifier as follows:

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Mode stitch to SLAVED
Latitude control to 0°
Hemisphere switch to N
Compensation switch to OFF

b. Repeat steps b through f of 4.6.4.14.1.

c. Rotate the displacement gyroscope heading transmitter at a rates of -6°, 30°, and 180° per minute.

(1) The ac voltage from connector 2J2-8 to 2J2-9 shall decrease to less than 0.5Vac within 30, 11, and 8 seconds, respectively.

(2) The dc voltage from connector 2J2-17 to 2J2-18 shall decrease to 0 ±1Vdc within 30, 11, and 8 seconds, respectively.

(3) The ac voltage from connector 2J2-8 to 2J2-9 shall increase to 6.5 ±1Vac within 18, 26, and 35 seconds, respectively, when the applied rates are removed.

(4) The dc voltage from connector 2J2-17 to 2J2-18 shall increase to a minimum of 4.0Vdc within 18, 26, and 35 seconds, respectively, when the applied rates are removed. (The azimuth detector input and the servoed repeater output shall be not less than 2° apart for requirement validity.)

d. Rotate the displacement gyroscope heading transmitter at

rates of -6°, -30°, and -180° per minute.

(1) The ac voltage from connector 2J2-8 to 2J2-9 shall decrease to less than 0.5Vac within 30, 11 and 8 seconds, repectively.

(2) The dc voltage from connector 2J2-17 to 2J2-18 shall decrease to 0 ±1Vdc within 30, 11, and 8 seconds, respectively.

(3) The ac voltage from connector 2J2-8 to 2J2-9 shall increase to 6.5 ±1Vac within 18, 26, and 35 seconds, respectively, when the applied rate is removed.

(4) The dc voltage from connector 2J2-17 to 2J2-18 shall increase to a minimum of 4Vdc within 18, 26, and 35 seconds, respectively, when the applied rate is removed. (The azimuth detector input and the servoed repeater output shall be not less than 2° apart for requirement validity.)

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4.6.4.14.1.2 Azimuth Oscillations - Azimuth oscillations shall be checked as follows:

a. Set the controls on the controller and the electronic control amplifier as follows:

Mode switch to SLAVED
Latitude control to 0°
Hemisphere switch to N
Compensation switch to OFF

b. Repeat steps b through f of 4.6.4.14.1.

c. With zero steady-state transmitter to produce an azimuth 1 second period.

(1) The ac voltage remain at 6.5 ±1Vac.

(2) The dc voltage from connector 2J2-17 to 2J2-18 shall remain greater than 8.0Vdc.

d. Repeat step c with an azimuth oscillation of ±2.0° with an 8 second period.

4.6.4.14.1.3 Steady-state Rate and Azimuth Oscillation - Steady-state rate and azimuth oscillation shall be checked as follows:

a. Set the controls on the controller and the electronic control amplifier as follows:

Mode switch to SLAVED
Latitude control to 0°
Hemisphere switch to N
Compensation switch to OFF

b. Repeat steps b through f of 4.6.4.14.1.

c. Drive the displacement gyroscope heading transmitter to produce an azimuth oscillation of ±0.80 with a 1 second period superimposed on a steady-state rate of turn of 300 per minute.

(1) The ac voltage from connector 2J2-8 to 2J2-9 shall decrease to less than 0.5Vac within 30 seconds.

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(2) The dc voltage from connector 2J2-17 to 2J2-18 shall decrease to O ±1Vdc within 30 seconds.

(3) The ac and dc voltages shall remain at these decreased levels until the steady-state rate of turn has been removed.

(4) The time required to return to normal condition shall be not more than 30 seconds.

4.6.4.14.1 ASK-44/A24G Electronic Control Amplifier - The roll erection and slaving control for the ASK-44/A24G shall be checked as follows:

a. Set the controls on the controller and the electronic control amplifier as follows:

Mode switch to SLAVED
Latitude control to 0°
Hemisphere switch to N
Compensation switch to OFF

b. Position the azimuth detector to O°.

c. Depress and hold the heading set control knob on the controller until the indicated servoed repeater heading becomes synchronized with the azimuth detector.

d. Reposition the azimuth detector to 3400.

e. Measure the dc voltage from connector 2J2-1 to 2J2-18. The voltage shall be greater than 8.OVdc.

f. Measure ac voltages-at the tabulated pins of connector 2J2. Voltages shall be within the following limits:

(2J2) Pins	Volts
8 to 9	6.5 ±1Vac
8 to 12	108.5 ±2Vac

g. Rotate the displacement gyroscope heading transmitter at a rate of 12° per minute for 1.5 minutes.

(1) The voltage from connector 2J2-8 to 2J2-9 shall remain at 6.5 ±1Vac.

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(2) The voltage from connector 2J2-17 to 2J2-18 shall remain greter than 8.0Vdc.

h. Repeat step g with a rate of -12° per minute for 1.5 minutes.

4.6.4.14.2.1 Switching Rate - The switching rate shall be checked as follows: (Note: During steps b through d, the dc voltage may temporarily become less than 1Vdc while the switching rate is applied. The time at which this occurs shall not be recorded.)

a. Set the controls on the controller and the electronic control amplifier as follows:

Mode switch to SLAVED
Latitude control to 0°
Hemisphere switch to N
Compensation switch to OFF

b. Repeat steps b through f of 4.6.4.14.2.

c. Rotate the displacement gyroscope heading transmitter at a rate of 240 and 1800 per minute.

(1) The voltage from connector 2J2-8 to 2J2-9 shall decrease to less than 0.5Vac within 12 and 4 seconds, respectively.

(2) The voltage from connector 2J2-17 to 2J2-18 shall decrease to 0 ±1Vdc within 12 and 4 seconds, respectively.

(3) The voltage from connector 2J2-8 to 2J2-9 shall increase to 6.5 ±1Vac within 8 and 14 seconds, respectively, when the applied rates are removed.

(4) The voltage from connector 2J2-8 to 2J2-9 shall increase to a minimum of 4.0Vdc within 8 and 14 seconds, respectively, when the applied rates are removed. (The azimuth detector input and the servoed repeater otput shall be not less than 4° apart for requirement validity.)

d. Rotate the displacement gyroscope heading transmitter at rates of -24° and -180° per minute.

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(1) The voltage from connector 2J2-8 to 2J2-9 shall decrease to less than 0.5Vac within 12 and 4 seconds, respectively.

(2) The voltage from connector 2J2-17 to 2J2-18 shall decrease to 0 ±1Vdc within 12 and 4 seconds, respectively.

(3) The voltage from connector 2J2-8 to 2J2-9 shall increase to 6.5 ±1Vac within 8 and 14 seconds, respectively, when the applied rate is removed.

(4) The voltage from connector 2J2-17 to 2J2-18 shall increase to a minimum of 4.0Vdc within 8 and 14 seconds, respectively, when the applied rate is removed. (The azimuth detector input and the servoed repeater output shall be not less than 4° apart for requirement validity.)

4.6.4.14.2.2 Azimuth Oscillations - Azimuth oscillations shall be checked as follows:

a. Set the controls on the controller and the electronic control amplifier as follows:

Mode switch to SLAVED
Latitude control to 0°
Hemisphere switch to N
Compensation switch to OFF

b. Repeat steps b through f of 4.6.4.14.2.

c. With zero steady-state rate, drive the displacement gyroscope transmitter to produce an azimuth oscillation of ±0.8° amplitude with a 1 second period.

(1) The voltage from connector 2J2-17 to 2J2-18 shall at 6.5 ±1Vac.

(2) The voltage from connector 2J2-17 to 2J2-18 shall remain greater than 8.0Vdc.

d. Disconnect all test equipment.

4.6.4 Control Assembly Operational Tests

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4.6.5.1 Test Connections - All control assembly components shall be connected in accordance with figure 6 using the simulated magnetic azimuth detector of 4.3.2.5. Connect a servoed repeater to the heading outputs at electronic control amplifier connector (2J1), and to the roll or pitch output at electronic control amplifier connector (2J4) as required. Connect 400 Hz input power to the following pins of connector 1J1 on the electronic control amplifier:

Neutral to 23
115V øA to 24
115V øB to 25
115V øC to 26

4.6.5.2 Special Test Conditions - Wait a minimum of 2 1/2 minutes after application of power before conducting any test. controls for the compass system controller and the electronic control amplifier shall be set as follows except where otherwise noted:

Mode switch to DG
Latitude control to local latitude
Hemisphere switch to local hemisphere
Compensation switch to OFF

4.6.5.3 Test Procedures - Unsatisfactory results that occur in areas not being tested at the time shall be investigated by performing the applicable individual test on the component involved. If heading drift trim adjustment is performed, the heading drift rate test for the displacement gyroscope shall be repeated- Azimuth rates greater than 6° per minute may cause removal of roll erection and slaving when testing a control assembly that uses the ASK-32/A24G-39. Azimuth rates greater than 24° per minute may cause removal of roll erection and slaving when testing a control assembly that uses the ASK-44/A24G. Sufficient time (up to 40 seconds maximum for the ASK-32/A24G-39 and up to 16 seconds for the ASK-44/A24G) shall be allowed to permit the control assembly to return to normal conditions after such rates cease. This may be particularly noticeable when observing the sync indicator pointer on the compass system controller.

4.6.5.3.1 Heading Rotational Polarity

4.6.5.3.1.1 DG Mode - The displacement gyroscope shall be rotated through a right-turn heading change of 90°. The transmitted heading signal from the electronic control amplifier shall increase by approximately 90°.

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4.6.5.3.1.2 SLAVED Mode - Turn the mode selector switch on the compass system controller to the SLAVED position and synchronize the control assembly by depressing and holding the heading set control knob on the compass system controller until the servoed repeater stops rotating. With the compass system controller having the timed automatic synchronization specified in 3.8.3.2, synchronization will take place automatically within 10 +2 seconds after switching to SLAVED mode. Note that the sync indicator pointer on the compass system controller is centered within ±1/16 inch. The procedure shall then be as follows:

a. Rotate the displacement gyroscope through a right-turn heading change of 600.

b. Rotate the azimuth detector through a right-turn heading change to produce a compass system controller sync indicator null. Null shall be achieved with a right-turn heading rotation of approximately 60 .

c. Hold the heading set control knob on the compass system controller depressed and rotate the azimuth detector through a right-turn heading change. The transmitted heading signal from the electronic control amplifier shall increase.

d. Turn the mode selector switch on the compass system controller to the DG position.

4.6.5.3.1.3 Compass or Emergency Mode - Turn the mode selector switch on the compass system controller to the COMP or EMERG position. The sync indicator pointer on the compass system controller shall become centered and the control assembly heading shall stabilize. The procedure shall then be as follows:

a. Rotate the displacement gyroscope through a turn of 60° ±10°. The control assembly heading shall not change.

b. Rotate the azimuth detector through right and left turns of 60° ±10°. The control assembly heading output shall follow the azimuth detector smoothly.

c. Turn the mode selector switch on the compass system controller to the DG position.

4.6.5.3.2 Roll Correspondence and Rotational Polarity - The displacement gyroscope shall be positioned to a right-wing-down roll angle of 30°. The transmitted roll signal from the electronic control amplifier shall be ±30° ±1°. The displacement gyroscope shall be returned to a level position.

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4.6.5.3.3 Pitch Correspondence and Rotational Polarity - The displacement gyroscope shall be positioned to a nose-up pitch angle of 30°. The transmitted pitch signal from the electronic control amplifier shall be 30° ±1°. The displacement gyroscope shall be returned to a level position.

4.6.5.3.4 Heading Set Function

4.6.5.3.4.1 DG Mode - The heading set control knob on the compass system controller shall be depressed and turned cw. The transmitted heading signal shall increase at a rate which is adjustable with low rate corresponding to small deflection angle.

4.6.5.3.4.2 SLAVED Mode - Turn the mode selector switch on the compass system controller to the SLAVED position. With the compass system controller having the timed automatic synchronization specified in 3.8.3.2, synchronization will take place automatically with 10 ±2 seconds after switching to the SLAVED mode. For models not having automatic synchronization, synchronize the control assembly by depressing and holding the heading set control. When the transmitted heading signal has stopped rotating, it shall remain stable while the heading set control knob on the compass system controller remains depressed. Release the heading set control knob. The following procedures shall then be accomplished:

a. Turn the mode selector switch on the compass system controller to the DG position.

b. Depress the heading set control knob on the compass system controller, deflect it cw for 1 to 3 seconds, then release it without returning it to center position.

c. Turn the mode selector switch on the compass system controller to the SLAVED position. The sync indicator pointer on the compass system controller shall show a positive (+) deflection. With the timed automatic synchronization function, the sync indicator pointer on the compass system controller will return to center position as the control assembly automatically synchronizes.

d. Turn the mode selector switch on the compass system controller to the DG position.

4.6.5.3.5 Slaving Rate and Correspondence - The slaving rate and correspondence test shall be conducted as follows:

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a. Turn the mode switch on the compass system controller to the SLAVED position.

b. Synchronize the control assembly, if necessary.

c. Rotate the displacement gyroscope through a 5° right turn- The sync indicator pointer on the compass system controller shall show full-scale positive (+) deflection.

d. Rotate the azimuth detector through a 10° right turn. The sync indicator pointer on the compass system controller shall show full-scale negative (-) deflection.

e. Desynchronize the control assembly and note the indicated heading.

f. Rotate the displacement gyroscope to change the heading 15° from the synchronized null position, and allow normal slaving to return the heading toward the null position. The time required for the heading to change from the 15° to the 5° off null position shall be between 5 and 10 minutes.

g. Turn the mode switch on the compass system controller to the DG position.

4.6.5.3.6 Control Assembly Heading Drift Rate - The displacement gyroscope shall be subjected to scorsby motion and the latitude control on the compass system controllers set to the local latitude. The control assembly heading shall be recorded and after a 1 hour interval shall be recorded again. The change in heading shall be ±1° maximum. (For the tests specified in 4.5.3.5 and 4.6.12 through 4.6.24, the normal limit shall be 1 1/2° maximum.)

4.6.6 Voltage and Frequency Variation - For this test and all remaining tests, interconnectIon of components, test equipment, power, and settings of control assembly controls shall be in accordance with 4.6.2 through 4.6.4 or 4.6.5, as applicable. The components shall be properly connected and operated at standard voltage and frequency, and set to a synchronized null with the simulated azimuth detector at a 0° heading. The category B voltage and frequency extremes of MIL-STD-704 shall be applied and the control assembly shall be resynchronized at each value. The shift in the synchronized null shall not exceed 3/4°. The components, as applicable, shall meet the tests specified in 4.6.2.3 with a 10 percent increase in tolerance, 4.6.2.5 with a 20 percent increase in tolerance, 4.6.4.7 with an input rate of 9 rpm and 4.6.5.3.6 with a 50° increase in tolerance. No damage shall result from the voltage and frequency extremes.

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4.6.7 Power Consumption - The control assembly shall give specified performance from a category B power source as defined in MIL-STD-704. The power required at nominal voltage and frequency shall not exceed the following values, exclusive of power supplied from the control assembly to operate associated external equipment.

a. ac power (three phase) 115/200V, 400 Hz.

(1) Displacement gyroscope: 75W, starting; 50W, running.

(2) ASK-321A24G-39 or ASK-44/A24G electronic control amplifier and the compass system controller: 50W.

b. ac/dc power: 5V

(1) Compass system controller lighting: 5.0W.

4.6.8 Vibration Error - The control assembly shall be operated in the SLAVED mode, synchronized to the simulated azimuth detector, and subjected to a vibration error test. During this test, the input vibration to the control assembly shall coincide with a power spectral density of 0.015 g^2/Hz for each frequency from 20 to 2,000 Hz. To accomplish this representation, the loading effect of the component under test shall be eliminated by proper load compensation. This random vibration shall be applied to the component along the three mutually perpendicular axes of the component. The signal source shall be either a prepared magnetic tape or a noise generator. To properly observe errors, the duration shall be from 10 to 15 minutes. The control assembly outputs of pitch, roll, and heading shall not vary by more than 3/4° from the initial position established with 10 percent of this vibration level, and shall return to the initial position when the vibration is again reduced to 10 percent of the test vibration level.

4.6.8.1 DG Mode - The control assembly shall be operated in the DG mode and subjected to the vibration error test. During this test, the input vibration to the control assembly shall coincide with a power spectral density of 0.002g^2/Hz for each frequency from 20 to 2,000 Hz. To accomplish this representation, the loading effect of the components under test shall be eliminated by proper load compensation. The random vibration shall be applied to the component along the three mutually perpendicular axes of each component. The signal source shall be either a prepared magnetic tape or a noise generator. The duration of the test shall be 15 minutes. The control assembly heading change shall not exceed ±1/20 when corrected for the earth's rotation.

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4.6.9 Low Field Strength - The control assembly shall be connected to a simulated azimuth detector and operated on a synchronized null. Excitation of the simulated azimuth detector shall be reduced to 102 ±2.5 mV 800 Hz to represent a horizontal field strength of 0.080 ±0.002 oersted. The simulated azimuth detector shall be oscillated through ±5° either side of null at 6 cpm. The control assembly heading shall be noted each minute after start of the oscillation until three consecutive readings are within 0.10 of each other. The difference between the initial control assembly heading and the final heading shall not exceed 0.5°. The test shall be run on three headings not less than 60° apart.

4.6.10 Slaved Dynamic Acuracy - The control assembly shall be operated in the SLAVED mode and synchronized to the simulated azimuth detector. The simulated azimuth detector shall be oscillated through ±5° either side of null at 6 cpm, and readings of the heading output shall be made at 1 minute intervals. Readings shall continue until three consecutive 1 minute readings are within 0.1° of each other, which shall occur within 45 q inutes from the start of the oscillation. The difference between the initial and final headings shall not exceed 1.0°.

4.6.11 Radio Noise Interference and Susceptibility - The radio noise and susceptibility test shall be conducted in accordance with MIL-I-6181. The control assembly shall be tested for radio noise interference over a frequency rangeof 150 kc to 400 mc and for susceptibility over a frequency range of 150 kc to 1,000 mc. During this test, the control assembly shall be properly connected with the displacement gyroscope operating on a 15° scorsby q echanism and with the lighting circuit in the compass system controller energized.

4.6.12 Temperature-altitude (Cycling) - The control assembly components shall be subjected to a temperature-altitude test in accordance with method 504, procedure I of MIL-STD-810 for Class 2 equipment. Maximum test temperature with control assembly operating shall be 71°C. Maximum non-operating temperature for the compass system controller shall be 85°C. Tests to determine satisfactory operation shall consist of 4.6.2.1; 4.6.4.14.1.1 (or 4.6.4.14.2.1), step d, except the -30° and -180° per minute rates shall not apply; and 4.6.5.3.5 with tolerances increased as noted. Tests to determine-performance characteristics shall consist of 4.6.2.1; 4.6.2.4; 4.6.4.14.1.1 (or 4.6.4.14.2.1), step c; 4.6.4.14.1.2 (or 4.6.4.14.2.2), step d; 4.6.4.14.1.3 and 4.6.5.3.6, as applicable, with the following tolerances increased. At the conclusion of this test, the control assembly components shall pass the individual tests with no tolerance increases.

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a. Test 4.6.2.1: 5 minutes will be permitted for completion of erection when at reduced temperatures.

b. Test 4.6.2.4: Limit is ±15° per hour.

c. Test 4.6.4.14.1.1: Time limits for decreasing voltages are 55, 15, and 10 seconds for rates of 6°, 30°, and 180° per minute, respectively. The time limits for increasing voltages after rates have ceased are 25, 37, and 40 seconds for rates of 6°, 30°, and 180° per minute, respectively.

d. Test 4.6.4.14.2.1: time limits for decreasing voltages are 18 and 5 seconds for rates of 24° and 180° per minute, respectively. The time limits for increasing voltages after rates have ceased are 13 and 16 seconds for rates of 24° and 180° per minute, respectively.

e. Test 4.6.5.3.6: Limit is ±2± per hour.

4.6.13 Humidity - The control assembly components shall be subjected to a humidity test in accordance with method 507, procedure I of MIL-STD-810. Upon completion, the displacement gyroscope shall meet the test specified in 4.6.2.1 and the other components shall meet the individual tests. There shall be no evidence of detrimental corrosion or rusting of external parts of the components.

4.6.14 Fungus - The compass assembly components shall be subjected to a fungus that in accordance with method 508, procedure I of MIL-STD-810. Upon completion of this test, there shall be no evidence of detrimental, deterioration, corrosion, or fungus growth on any external surface of the components.

4.6.15 Rain - The compass system controller and electronic control amplifiers shall be subjected to a rain test in accordance with method 506, procedure I of MIL-STD-810. At the conclusion of this test, the compass system controller and electronic control amplifiers shall meet the tests specified in 4.6.4.7. There shall be no evidence of water damage. In addition, the moisture-sealed compass system controller shall be mounted with its lighting panel facing up and subjected to the Waterproofness Test of RTCA Document No. DD160, Waterproofneas Test 10.2. At the conclusion of the test, the outside surface of the compass system controller shall be wiped dry, the access cover removed, and the interior inspected. There shall be no evidence of damage. The cover shall be reattached and the compass system controller shall be subjected to the tests of 4.6.3.

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4.6.16 Salt Fog - The control assembly components shall be subjected to a salt fog test in accordance with method 509, procedure I of MIL-STD-810. Upon completion of this test, the components shall meet the tests specified in 4.6.2.1; 4.6.4.7; 4.6.4.14.1.1 (or 4.6.4.14.2.1) step d, except the -30° and -180° shall not apply; and 4.6.5.3.5, as applicable. There shall be no evidence of corrosion that would shorten the operational life of the components.

4.6.17 Dust – The compass system controller and electronic control amplifiers shall be subjected to a dust test in accordance with method 510, procedure I of MIL-STD-810. Upon completion of this test, the compass system controller and electronic control amplifiers shall meet the tests specified in 4.6.4.7, and 4.6.4.14.1.1 (or 4.6.4.14.2.1), step d, except the -30° and –180° shall not apply. There shall be no evidence of damage due to this test.

4.6.18 Temperature Shock – The control assembly components shall be subjected to a temperature shock test in accordance with MIL-STD-810, method 503. Upon completion of this test, the components shall meet the tests specified in 4.6.2.1, 4.6.4.7, 4.6.4.14.1.1 (or 4.6.4.14.2.1), step d, except the -30° and –180° rates shall not apply; and 4.6.5.3.5.

4.6.18.1 Leak Rate - The displacement gyroscope seal shall not permit the equivalent of more than 10 percent loss of the filling medium after 1,000 hours at a pressure differential equal to a 100,000 foot altitude.

4.6.19 Vibration Failure - The components shall be subjected to a vibration failure test in accordance with MIL-STD-810, method 514.1, equipment category (a), curve M, figure 514.1-1. At the conclusion of this Lest, the components shall meet the following tests:

a. Compass System Controller - - - - - - - - - 4.6.3
b. Displacement Gyroscope - - - - - - - - - - 4.6.2
c. Electronic Control Amplifier - - - - - - 4.6.4

4.6.20 Acoustical Noise – The control assembly shall be operated in the SLAVED mode, synchronized to the simulated azimuth detector, and subjected to an acoustical noise test in accordance with MIL-STD-810, method 515, intensity category A, procedure I. The control assembly outputs of pitch, roll, and heading shall not vary by more than 3/4° from the initial position established with 10 percent of this noise level, and shall return to the initial position when the noise level is again reduced to 10 percent of the test noise level.

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4.6.21 Explosion - The non-hermetically sealed control assembly components shall be subjected to an explosion test in accordance with MIL-STD-810, method 511, procedure I.

4.6.22 Shock - The control assembly components shall be subjected to a shock test in accordance with M1L-STD-810, method 516, procedure 1, figure 516.1-1 for flight vehicle equipment, except the shock force shall be 15g in lieu of 20g, and procedure III. Upon completion of this test, the components shall individual tests.

4.6.23 Acceleration - The control assembly components shall be subjected to an acceleration test in accordance with MIL-STD-810, method 513, procedures I and II, for airplanes and helicopters. If procedure II is conducted on a centrifuge, the displacement gyroscope need not be subjected to UP and DOWN accelerations. Performance tests shall not be conducted during the acceleration but the outputs shall be monitored during procedure II. Upon completion of procedure 1, the control assembly shall meet sampling plan A tests except the individual tests need not be conducted. Upon completion of procedure 11 testing, the control assembly shall meet sampling plan A tests.

4.6.24 Reliability - The control assembly components shall be properly connected as shown in figure 6 and operated to assure proper setup. The control assemblies or the components shall be placed in test chambers which meet the environmental cyclic conditions of test level F of MIL-STD-781, except that the temperature extremes shall be +55°C and -44°C, and vibration shall be 0.5g. After each 200 hours of equipment operating time the components shall meet the appropriate individual tests and the control assemblies shall meet the sampling plan A tests. See 4.4.1.1 for accept-reject criteria.

4.6.24.1 Longevity - At the completion of the reliability test specified in 4.6.24, three control assemblies or components shall continue to be tested for a total of 2,000 hours each per 3.5.3.1.

4.6.24.2 Allowances and Failures - The increase in tolerances allowed for reliability testing shall be as shown in table XXII. out of tolerance calibration-type requirements will only be considered as failures if they occur in two successive cycles. Except for the above reservations, a failure shall be as defined in 6.3.2.

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TABLE XXII. Reliability Testing Tolerances

Percent Increase in Tolerance	Reliability Test Time Per Control Assembly or Component
0	0 to 500 hrs
25	500 to 1,000 hrs
50	1,000 to 1,500 hrs
100	1,500 hours and over

5. PACKAGING

5.1 Preservation and Packaging - Sealed control assembly components shall be preserved in accordance with MIL-P-116, method 1A and unsealed control assembly components shall be preserved in accordance with method II.

5.2 Packing - Packing shall be in accordance with MIL-STD-794, level A for export shipment and level B for domestic shipment.

6. NOTES

6.1 Intended Use - The attitude gyroscope control assemblies covered by this specification are intended for use in either fixed-wing or rotary-wing aircraft to provide pitch, roll, and heading displacement signals of the aircraft for indication and tie-in to other equipment. Provisions are made for various types of operation and error compensation, depending upon the application and associated equipment.

6.2 Ordering Data

6.2.1 Procurement Requirements - Procurement documents shald specify the following:

a. Title, number, and date of this specification.

b. Type of Electronic Control Amplifier and type of Compass System Controller required (see 3.2) for the aircraft application.

c. Disposition instructions for test samples (see 4.4.1)0

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d. When sampling plan C tests will not be conducted (see 4.5.2.3).

e. Levels of packaging and packing required (see 5.2).

f. Specify control assembly components to be tested (see 3.2).

g. Specify when a First Article testing is required for inspection approval (see 3.1, 4.4 and 6.2.2.1).

h. Specify when reliability demonstration tests are not to be performed (see 4.4.1.1).

6.2.2 Data Requirements - When thi specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.5.3.1	Reliability Assurance Production Acceptance Phase Test Results	UDI-T-21350
4.5.3.2	Reliability Assurance Test Procedures	UDI-T-21347

(Copies–of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

a. Specify data requirements in accordance with DD Form 1423.

6.2.2.1 First Article - When a first article is required by the contract (see 6.2.1), it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be a sample selected from production items as specified in 4.4. The first article should consist of 3 samples as specified in 4.4.1. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

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6.3 Definitions

6.3.1 Hermetic Seal - A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic materials, or both. A hermetic seal is not intended to included seals accomplished by gaskets or O-rings.

6.3.2 Failure - where a condition is specified as a requirement, a failure will be considered as any electrical or mechanical condition which fails to meet the requirements specified herein.

6.4 Changes From Previous Issue - Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.5 Government-Furnished Property - The contracting officer should arrange to furnish the property listed in 3.2.2.

Preparing activity:
Navy - AS

(Project 6610-NO98)

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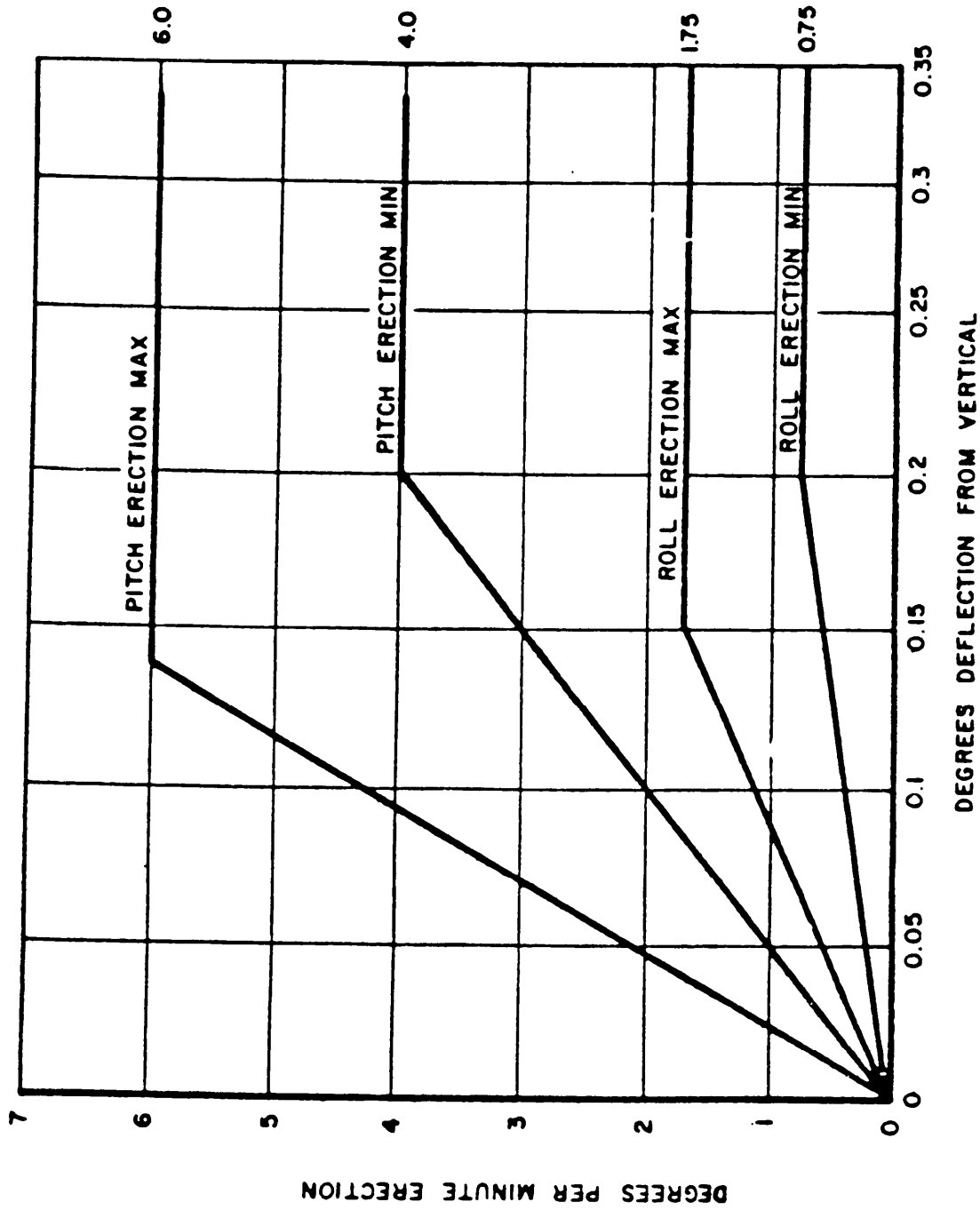


FIGURE 1. ERECTION RATE AS A FUNCTION OF DEFLECTION

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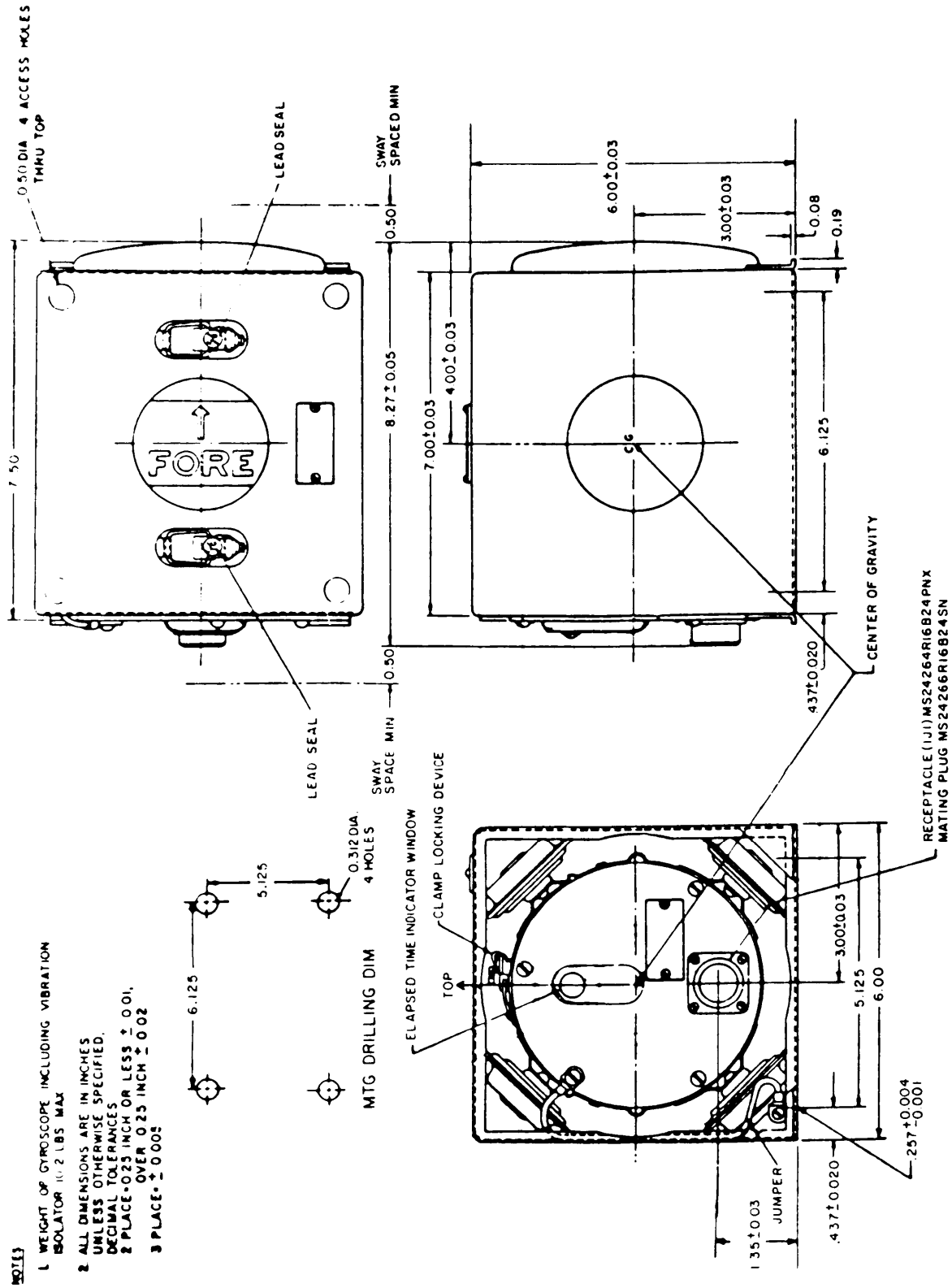


FIGURE 2. DISPLACEMENT GYROSCOPE SBK-11A/A24G-26

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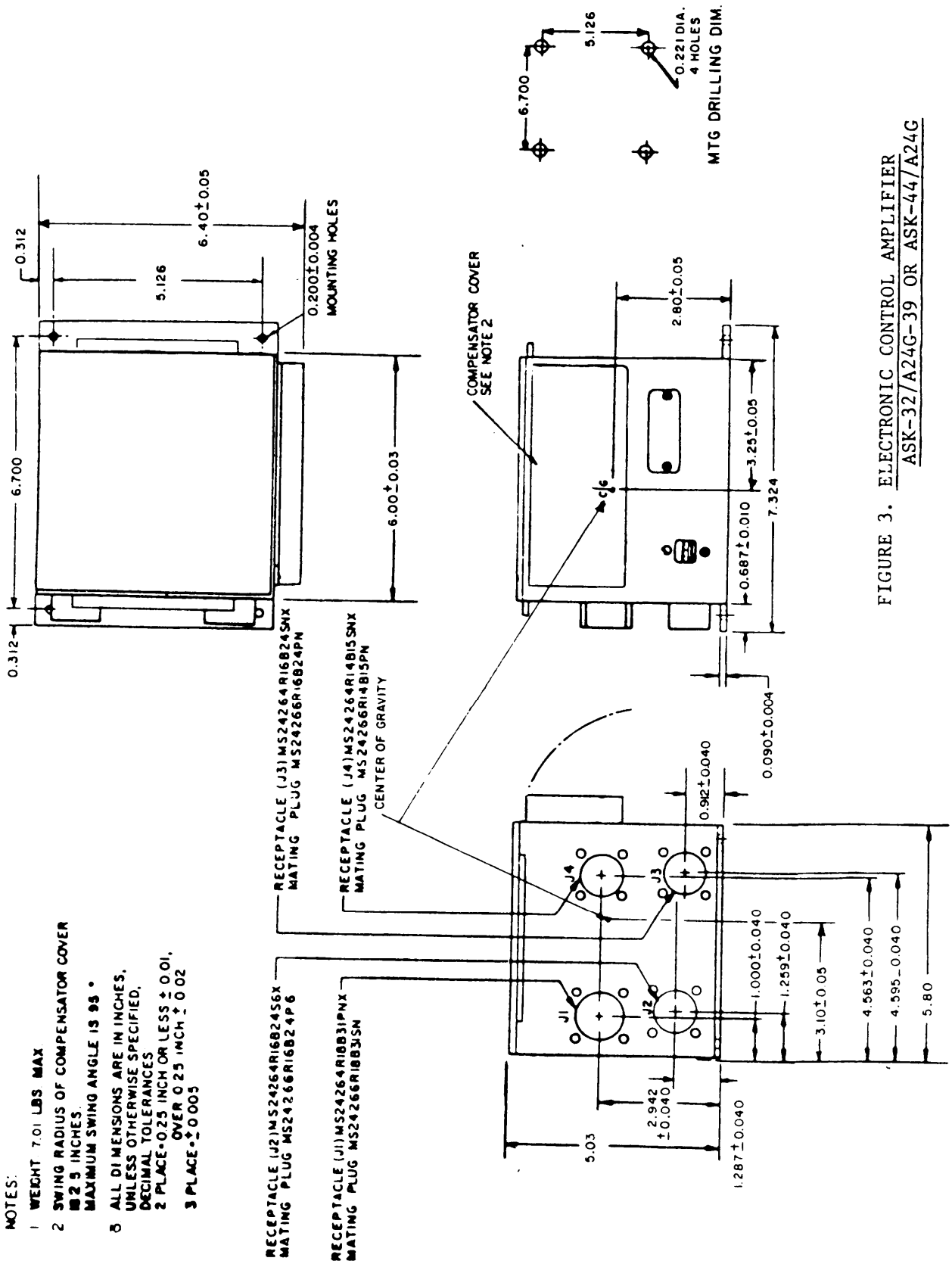


FIGURE 3. ELECTRONIC CONTROL AMPLIFIER ASK-32/A24G-39 OR ASK-44/A24G

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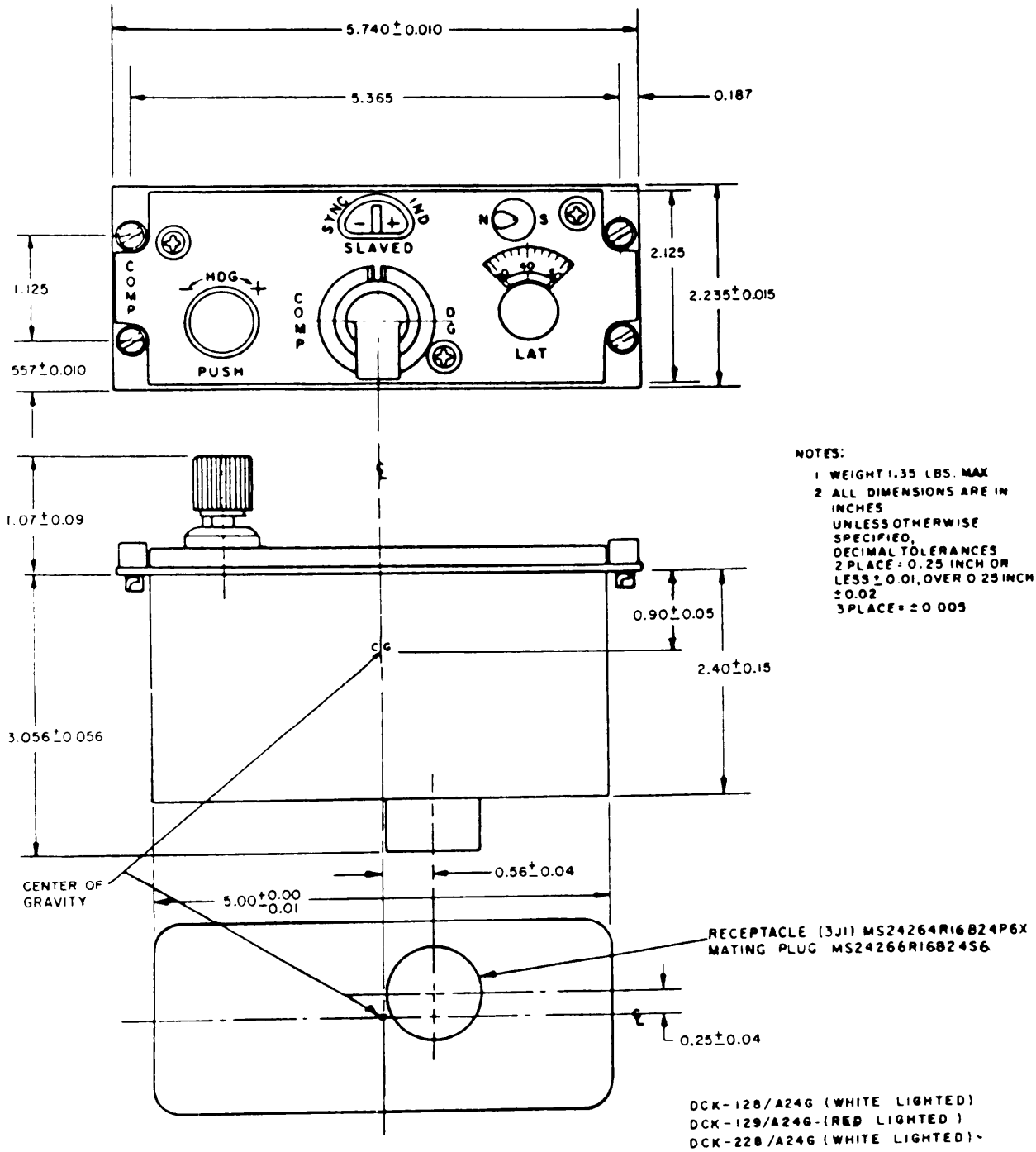
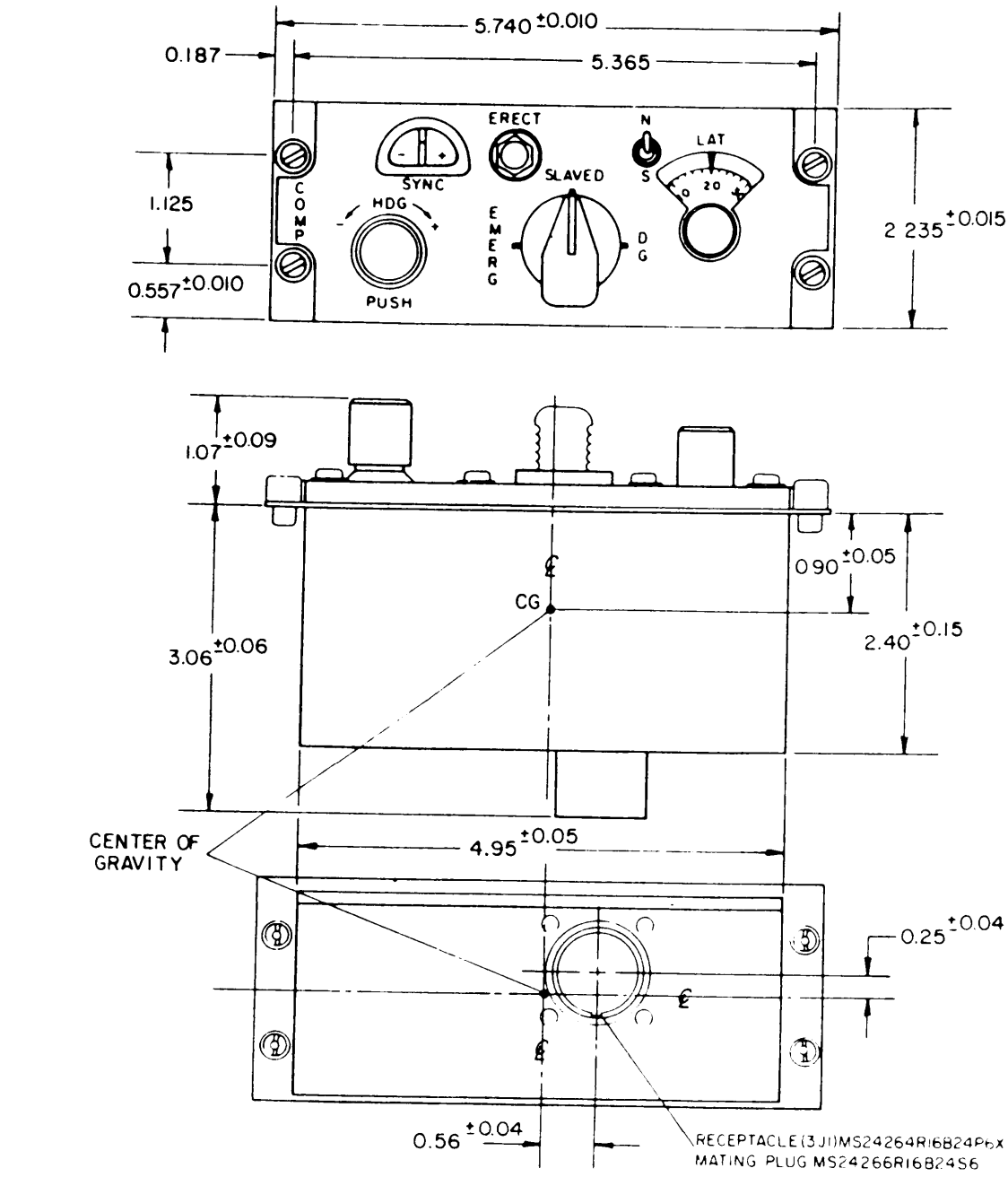


FIGURE 4. COMPASS SYSTEM CONTROLLER

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NOTES:
1. WEIGHT 1.5 LBS MAX
2. ALL DIMENSIONS ARE IN INCHES
   UNLESS OTHERWISE SPECIFIED, DECIMAL TOLERANCES:
   2 PLACE = 0.25 INCH OR LESS ±0.01, OVER 0.25 INCH ±0.003
   3 PLACE = ±0.003

DCK-234/A24G (WHITE LIGHTED)
DCK-235/A24G (RED LIGHTED)
DCK-236/A24G (WHITE LIGHTED)

FIGURE 5. MOISTURE-SEWED COMPASS SYSTEM CONTROLLER

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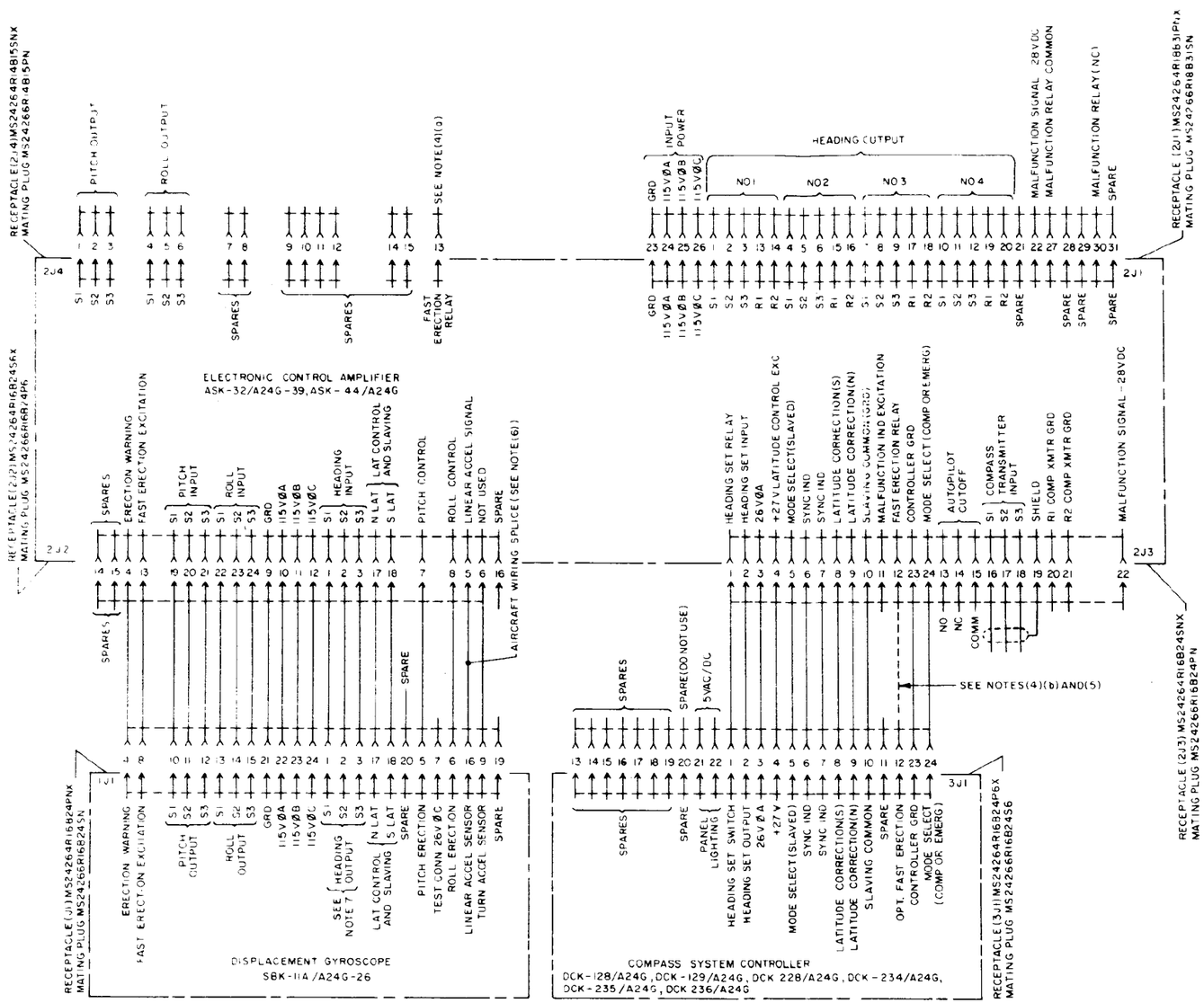


FIGURE 6. A/A24G-39 ATTITUDE GYROSCOPE CONTROL ASSEMBLY, INTERCONNECTION WIRING DIAGRAM

NOTES

(1) INSTALLATION AND TESTING SHALL BE IN ACCORDANCE WITH E I 590 AND ET590.

(2) AIRCRAFT WIRING SHALL BE IN ACCORDANCE WITH MIL SPEC MIL-W-5088.

(3) WHEN NECESSARY, RIGHT ANGLE CONNECTORS MAY BE USED IN LIEU OF STRAIGHT CONNECTORS.

(4) FAST ERECTION FOR DCK-128/A24G, DCK-129/A24G, DCK-228/A24G (USE ONLY ONE METHOD).

(a) PREFERRED METHOD:

1. PROVIDES MANUAL CONTROL OF FAST ERECTION INDEPENDENT OF HEADING SYNCHRONIZATION WHEN AN EXTERNAL NORMALLY OPEN SPRING-RETURN SWITCH IS CONNECTED BETWEEN 2J4, PIN 13, AND POWER GROUND.

(b) OPTIONAL METHOD:

1. PROVIDES MANUAL CONTROL OF FAST ERECTION BY DEPRESSING HDG/PUSH KNOB ON COMPASS SYSTEM CONTROLLER WHEN AN AIRCRAFT JUMPER WIRE IS ADDED FROM 2J3, PIN 12 TO 3J1, PIN 12.

(5) FAST ERECTION FOR DCK-234/A24G, DCK-235/A24G, DCK-236/A24G.

(a) ADD AIRCRAFT JUMPER WIRE 2J3, PIN 12 TO 3J1, PIN 12. (THIS PROVIDES CONTROL AT THE ERECT SWITCH FOR THESE COMPASS SYSTEM CONTROLLERS).

(6) WITH AN ASK-44/A24G ELECTRONIC CONTROL AMPLIFIER INSTALLED, A REDUCED PITCH ERECTION RATE (DURING FLIGHT) OPTION CAN BE UTILIZED, IF THIS POINT IS REFERENCED TO GROUND THROUGH A RELAY OR OTHER SWITCHING DEVICE THAT IS ACTIVATED WHEN AIRCRAFT IS AIRBORNE.

(7) LEAD S2 INTERNALLY CONNECTED TO GROUND IN DISPLACEMENT GYROSCOPE.

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a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

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